
Invent a Dataset: Measuring dataset generation abilities with zero seed data

Shivalika Singh^{*1}, Andrija Djurisc^{*1}, Gbemileke Onilude¹, Sudip Roy¹,
and Sara Hooker¹

¹Adaption

Building datasets remains one of the most manual and brittle parts of AI development. In this technical report, we focus on the most extreme but also most prevalent setting real world practitioners face: a zero data regime. Here, practitioners don't have any data for the capability they want to learn. We introduce INVENT-A-DATASET which is a prompt based system to go from dataset description to realistic and large scale post-training datasets. We evaluate INVENT-A-DATASET against five frontier model APIs including Anthropic , Google , Open AI , DeepSeek , Zai . Across eight task types and dataset sizes up to 20K samples, INVENT-A-DATASET significantly outperforms with both the highest quality (17% relative gains) while simultaneously producing the most diverse samples (19% relative gains). Its diversity advantage widens with scale of training dataset size (from parity at 200 samples to 37% relative gains at 20K samples). This translates into considerable downstream training gains, resulting in far more performant post-trained models. Invent API fine-tune consistently ranks higher compared to other generator fine-tunes across different post-trained model architectures.

website: <https://adaptionlabs.ai/invent-a-dataset>

1 Introduction

Datasets are the raw materials that fuel breakthroughs. Progress at the AI frontier increasingly depends on high-quality training data that continues to challenge models (Kulikov et al., 2026). Lack of access to data is also one of the biggest determinants of who gets to shape the frontier of AI (Longpre et al., 2024). However, securing high quality data and sufficient volume to target new model capabilities remains incredibly challenging. Human-annotated training datum are expensive to collect and have historically been static representations of the world (Kulikov et al., 2026; Salazar et al., 2026; Singh et al., 2025; 2024a; Boubdir et al., 2023b). It takes considerable time to collect additional data, which results in datasets very far from the rich, ever-evolving environment we navigate as humans (Roh et al., 2019).

In this technical report, we focus on the most extreme end of this data divide. **A zero training data regime where no data exists for a desired capability.** This is a setting we take seriously because it is also the most common setting real world practitioners face. Several solutions have been proposed for this zero data regime, but fail to guarantee both diversity and quality of the final dataset. Dataset retrieval agents use internet search to comb for relevant datasets but this

^{*}Joint first authors.

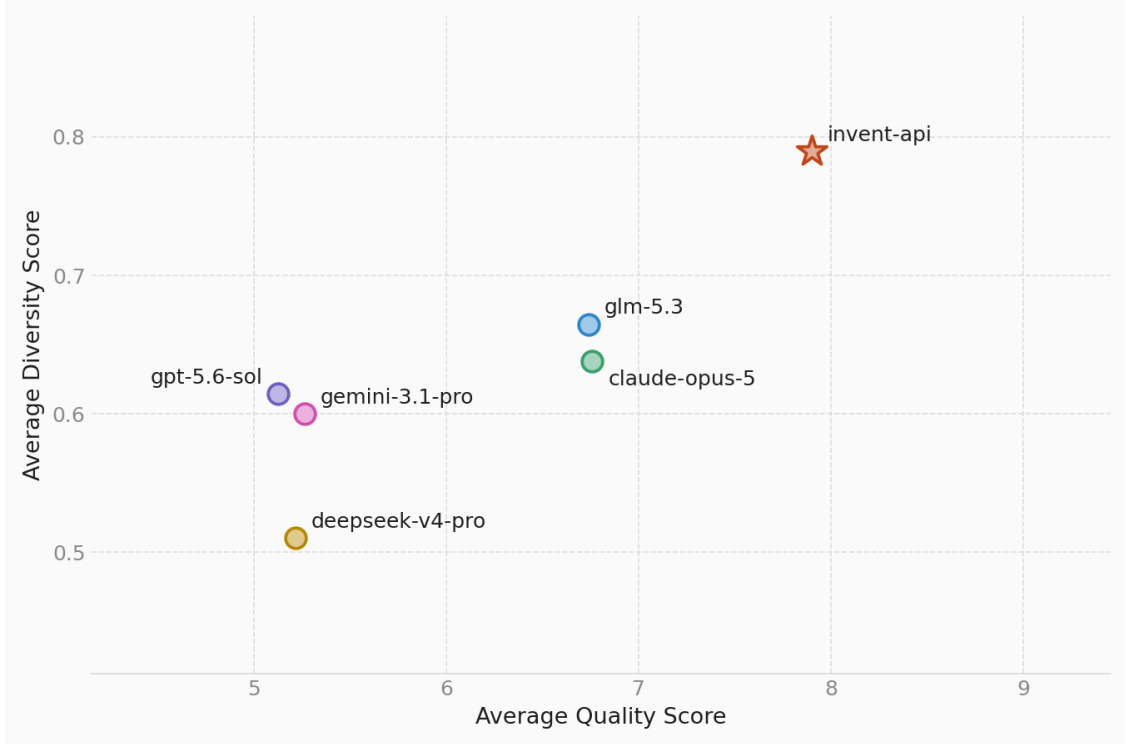


Figure 1: **Diversity vs. quality trade-off.** Quality (x-axis) versus diversity (y-axis), measured by DCScore with $\tau = 0.05$, for four external models and the Invent API. Both scores are averaged over eight 5K-sample datasets in unconstrained setting. Higher is better on both axes; the upper right is optimal. The Invent API attains high quality and high diversity simultaneously.

doesn’t guarantee returning datasets that are formatted correctly for AI training (Lin et al., 2025; Chapman et al., 2020b; Li et al., 2025a). Retrieved datasets are also unlikely to satisfy all the unique constraints a practitioner may have such as formatting, language coverage, output tone. This limits the feasibility of introducing new properties, or explicitly optimizing for task-specific metrics.

Alternatively, practitioners often use existing models to generate synthetic data related to their capability description. However, most synthetic data techniques assume that you either start with a seed pool of prompts or already have a dataset to improve (Wang et al., 2023a; Xu et al., 2024). Without these seed prompts, a significant challenge is scaling to the dataset sizes required for successful post-training while preserving corpus diversity (Dohmatob et al., 2024; Briesch et al., 2023; Shumailov et al., 2023; Bertrand et al., 2024; Guo et al., 2024). Large Language Models (LLMs) frequently suffer from low output diversity (Holtzman et al., 2019; Xu et al.). In the absence of diverse prompts, models with standard maximization-based decoding are known to get stuck in redundant consecutive repetitions. Furthermore, distilling synthetic data for the purpose of training is frequently restricted by licensing: many proprietary state-of-art models have terms of service that explicitly restrict training on their outputs (Wiggers, 2025).

In this work, we propose INVENT-A-DATASET to solve for several of these pain points. INVENT-A-DATASET enables autonomous, demand-driven data curation that goes from a single dataset description to realistic and AI training ready data (Adaption Labs, 2026). We release INVENT-

A-DATASET with permissive licensing that explicitly allows for downstream commercial use in training. We evaluate INVENT-A-DATASET against directly prompting five frontier model APIs including Anthropic, Google, Open AI, DeepSeek, Zai at diverse dataset sizes up to 20K samples, measuring quality, diversity, and dataset relevance. Our findings can be listed as follows:

1. Invent API significantly outperforms with both the highest quality (17% relative gains in quality against the strongest baseline) while simultaneously producing the most diverse samples (19% at 2k).
2. This consistent advantage in diversity widens with scale. Unlike most baselines, whose diversity stagnates or declines as datasets requests grow from 2K to 20K samples (−15% for Claude-Opus-5, −8% for GLM-5.3, −7% for GPT-5.6-sol, and +0.4% for DeepSeek-V4-pro), Invent API becomes more diverse with scale (+7%) while maintaining stable quality.
3. Adding constraints to the dataset request like tone and output format reduces the diversity of outputs by 15% to 22% for every provider we evaluate. In contrast, Invent API maintains its diversity for both constrained and the harder unconstrained queries, outperforming the second best, GLM-5.3, by a remarkable 58% at 20K for constrained queries.
4. These quality gains translate to downstream post-training quality. We independently post-train and rank the fine-tuned models’ generations across the different data variants. Invent API fine-tune ranks first on 54% of the test prompts for Llama-3.3-70B and 41% for Gemma-4-31B-it, outperforming both the un-tuned base model and the variants fine-tuned on Claude-Opus-5 and GLM-5.3 generated data in each case.

Our results suggest it is now feasible to automatically target new capabilities from a zero data regime. Historically, the high cost of collecting and curating data has precluded adapting training sets “on-the-fly” to increase coverage or task diversity. INVENT-A-DATASET is an important step toward continuously optimized training sets which can evolve with tasks and target capabilities.

2 Experimental Setup

2.1 Evaluation Setup

Invent is a dedicated API end-point to generate diverse datasets from a prompt. We benchmark Invent API against a variety of proprietary and open-weights models. Across each model, desired dataset size and dataset query, we measure the ability to generate a high quality and diverse dataset. We evaluate eight representative dataset queries chosen to represent a range of **1)** task types, **2)** languages and **3)** domains ranging from customer support to medical QA. Additionally, for five of these datasets we compose an even harder version with additional formatting constraints such as tone constraints, json format and multiple choice output restrictions. These are designed to reflect popular practitioner requirements for training datasets (Zhou et al., 2023; Jiang et al., 2024; Wen et al., 2024). We refer to these two sets of evaluation queries as **unconstrained** and **constrained** queries respectively. The evaluation queries for three datasets are included in Table 8 and remaining in Appendix A.3.

Models Considered. Our benchmark set includes proprietary models such as Anthropic’s Claude Opus 5 (Anthropic, 2026), Open AI’s GPT-5.6 Sol (OpenAI, 2026), Google’s Gemini-3.1-pro

Dataset	Unconstrained Query	Constrained Query
African QA	Dataset containing question-and-answer pairs covering diverse topics related to African research, literature, history, and current events.	Dataset containing multi-turn conversations between a teacher and a student on an exam , covering diverse topics related to African research, literature, history, and current events. Prompts must use OpenAI format while responses should be plain text.
Legal QA	Dataset containing question-answer pairs focusing on legal advice, factual explanations, and comparative analysis across diverse topics such as copyright law, government types, and liability.	Dataset containing question-answer pairs focusing on legal advice, factual explanations, and comparative analysis across diverse topics such as copyright law, government types, and liability. Questions should offer four choices; answers are limited to A, B, C, or D with no additional text.
Serbian Car Ads	A dataset containing a collection of used car advertisements in Serbian, featuring detailed descriptions of vehicle conditions, service histories, and equipment packages.	A dataset containing a collection of summaries of used car advertisements in Serbian, featuring detailed descriptions of vehicle conditions, service histories, and equipment packages. Each prompt is an ad summary, and each response is a JSON object containing vehicle_description, service_history, and equipment_packages.

Table 3: The single request queries used to generate datasets. Queries are chosen to represent a range of 1) task types, 2) languages and 3) domains. We evaluate on both **unconstrained** and **constrained** versions of the query, where the constrained query requires additional requirements such as 1) tone constraints, 2) json format, 3) multiple choice output restrictions. Differences in the constrained query are **highlighted**.

(Google DeepMind, 2026). For the proprietary APIs we choose, it is against their terms of service to use data outputs for training. However, we include these models for an evaluation purpose as an upper bound in performance that is meaningful to calibrate against. Additionally, many practitioners appear to rely on these models for synthetic data generation despite it being prohibited. We also include open-weight models such as Z-ai’s GLM-5.3 (Z.ai, 2026) and Deepseek AI’s DeepSeek-V4-Pro-0813 (DeepSeek-AI, 2026) whose license permits use of outputs for downstream training. For our experiments, we use these models via the OpenRouter API (OpenRouter, 2026).

Generation configurations. We kept all model hyperparameters at their default values (temperature 1.0 where possible to specify, reasoning disabled). INVENT-A-DATASET returns a dataset in a single call, however in practice we observed that other models would rarely return the required number of samples if only a single call was used. At scale, single call requests often led to significant diversity collapse. While we continued to benchmark INVENT-A-DATASET using only a single call, we evaluated a set of different approaches on the other models benchmark : (i) *one-by-one*, prompting the model to produce a single sample per request (in contrast to requesting entire dataset in one query) (ii) *batch of 100*, producing samples in fixed batches of N=100 (iii) *max batch size*, producing as many samples as possible in a single request depending on model’s output token limit which was set to 128K across all models. As shown in Figure 2, diversity collapses almost entirely (≈ 0.00 – 0.03 across all four models) for the *one-by-one* approach, while *max batch* strategy is consistently the most diverse. Based on these results, we selected *max batch* as the default generation approach.

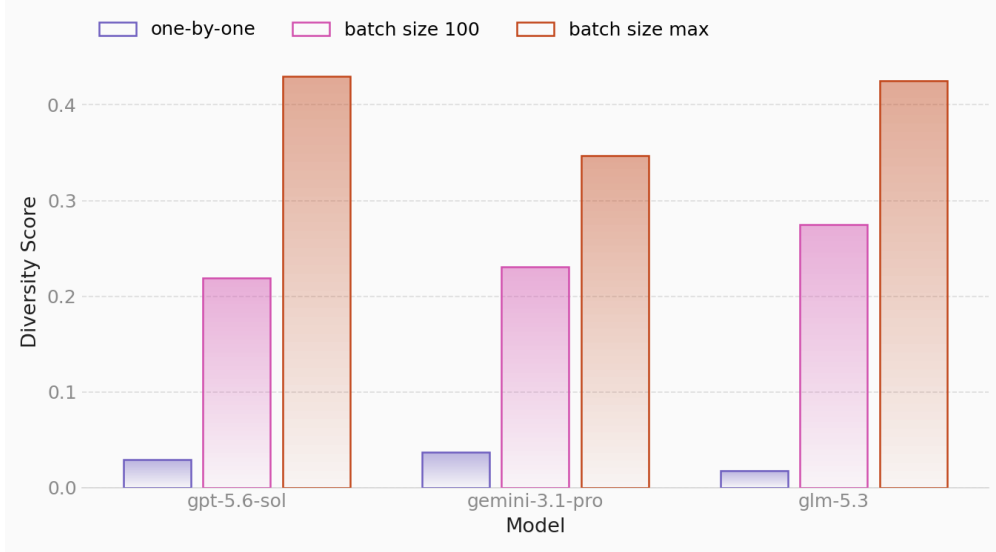


Figure 2: **Diversity across different generation strategies.** We compare strategies for generating a dataset from a single query request (Medical QA) across three models: Gemini 3.1 Pro, GPT-5.6 Sol, and GLM-5.3. Strategies differ in how many samples are produced per request: one sample at a time (*one-by-one*), fixed batches of 100, or as many samples as fit within the model’s maximum output-token limit (*batch_size max*). The y-axis shows DCScore ($\tau = 0.1$), where higher is better. Diversity generally increases with batch size, and generating samples one by one leads to near-complete diversity collapse (≤ 0.03) for every model.

We adopt this for all subsequent experiments, unless otherwise specified.

Dataset scaling. In practice, a core goal of obtaining sufficient data is to train a model. Hence, we are interested in the ability to generate data at a sufficient scale that is viable for training. For each query and model generator we consider, we invent data at four distinct scales: 200, 2000, 5000, 20000 samples. Larger scales allows us to measure how quality and diversity is impacted as the practitioner requests larger datasets for post-training purposes.

Downstream post-training set-up. To test whether using Invent API translates into downstream model performance gains, we repeat the same procedure independently for three models (Llama-3.3-70B-Instruct, Gemma-4-31B-it and Qwen3.5-9B) corresponding to different model families and model sizes. We post-train each of them on the 20K **Medical QA** dataset variants produced by every generator (samples shown in Table 5) with LoRA SFT. Rather than hand-tuning a training recipe, we use Autoscientist (Adaption Research Staff, 2026) to run a hyperparameter search on one of the **Medical QA** data variants. To control for the impact of dataset alone, the configuration proposed by Autoscientist is then applied to every generator’s dataset. As a result, every model in the comparison starts from the same base and is trained with the same optimizer settings, for the same number of epochs, on the same number of samples, so the training data is the only variable that differs between them.

API	Prompt	Completion
Claude Opus 5	“How is hypertension diagnosed, and what blood pressure numbers matter?”	“Hypertension is diagnosed from properly measured, repeated blood pressure readings rather than a single office value. The patient should be seated quietly for five minutes, back supported, feet flat, [...]”
Gemini 3.1 Pro	“What class of anticoagulant is Dabigatran?”	“Dabigatran is a Direct Thrombin Inhibitor (DTI). It reversibly binds to the active site of thrombin (Factor IIa), preventing the conversion of fibrinogen to fibrin. It was the first DOAC approved as an [...]”
DeepSeek V4 Pro	“What is the mechanism of action of digoxin?”	“Digoxin inhibits the ATPase pump in cardiac myocytes, raising intracellular sodium and, via the sodium-calcium exchanger, intracellular calcium [...]”
GPT 5.6 Sol	“What are warning signs of a heart attack?”	“Warning signs include pressure, squeezing, heaviness, or pain in the chest, sometimes spreading to an arm, shoulder, jaw, back, or upper abdomen. Shortness of breath, sweating, nausea, dizziness, or unusual fatigue may also occur, and symptoms can be atypical in women, older adults, and people with diabetes. [...]”
GLM 5.3	“What are the diagnostic criteria for hypertension?”	“Hypertension is diagnosed based on repeated elevated office blood pressure measurements. Under the 2017 ACC/AHA guidelines, stage 1 is systolic 130–139 or diastolic 80–89 mmHg, and stage 2 is 140/90 mmHg or higher. Out-of-office ambulatory or home monitoring is recommended to confirm the diagnosis. [...]”
Invent API	“A 62-year-old male presents with progressive shortness of breath, bilateral inspiratory crackles, and a dry cough. He has a 30-year history of asbestos exposure. Discuss the differential diagnosis between idiopathic pulmonary fibrosis and asbestosis.”	“ Differential Diagnosis: IPF vs. Asbestosis. The presentation of progressive dyspnea, dry cough and bilateral ‘Velcro’ crackles with a 30-year asbestos history is a classic diagnostic challenge. Both are chronic, progressive fibrosing interstitial lung diseases sharing a Usual Interstitial Pneumonia (UIP) pattern. [...]”

Table 5: Example prompt–completion pairs generated by each API for the **Medical QA** dataset, whose query was: “*Dataset of medical question-answer pairs covering diagnoses, treatments, drug explanations, and healthcare roles. Each entry includes a prompt and a detailed, expert-generated response*”

2.2 Evaluation Metrics

Diversity Metrics. We measure the generated dataset diversity along two complementary axes i.e. *semantic* and *lexical*.

- **Semantic diversity.** As our primary, semantically grounded measure we use the DCScore (Zhu et al., 2025). The DCScore is an embedding based metric that treats diversity evaluation as a sample classification task, enabling the metric to capture mutual relationships among

the samples. We use the Qwen-3-4B-embedding model (Zhang et al., 2025) as our embedding model to compute diversity scores since it supports multiple languages and 32K context length which is essential to correctly evaluate the datasets in our benchmark suite.

- **Lexical diversity.** In addition to DCScore, we also report metrics related to textual form rather than semantic content computed on the generated *prompts*: (i) *n*-gram diversity (NGD), the summed ratio of unique to total *n*-grams for $n \in \{1, \dots, 4\}$ over the corpus (Shaib et al., 2026; Padmakumar & He, 2024), where higher is more diverse; (ii) *compression ratio* (CR), the gzip ratio of raw to compressed corpus size (Shaib et al., 2026), where higher indicates more redundancy; and (iii) *exact-duplicate rate*, the fraction of prompts that are verbatim repeats.

For all these metrics, we report them on a fixed 2,000-prompt random sample from each generated corpus when comparing across dataset scales .

Quality Metrics. We adopt a rubric-based LLM-as-a-Judge methodology to evaluate quality, as it has been to shown by several works to be correlated with human judgement (Zheng et al., 2023; Liu et al.).

- **Prompt and completion quality** is scored on a scale of 0–10. Detailed description of these metrics and full judging prompt (see A.1) along with per-dataset results at various dataset size scales (see A.2) have been reported in the Appendix.
- **Relevance** which measures how well generated samples address the underlying intent and scope of the user’s request, also scored on the scale from 0 to 10. Relevance score validates that a sample matches the request’s task type and subject matter, is written in the target language and follows the constraints the request explicitly states (e.g. length, format, structure, style).
- **Post-training quality** We employ LLM-as-a-Judge ranking to compare models fine-tuned on Medical QA data from each generator. These are evaluated on a set of in-house, held-out set of medical domain prompts. We don’t derive our test set from any single generator’s data since such a split would be drawn from that generator’s own distribution and would be biased towards that model (Torrallba & Efron, 2011; Teney et al., 2020). Our test prompts are instead sampled from the real-world distribution of medical queries and are held out from every generator’s training data. This is a harder, out-of-distribution test and a more faithful indicator of downstream generalization (Recht et al., 2019; Koh et al., 2021). For each test prompt, the judge (Gemini 3.1 Pro) sees all responses in a single call and returns a best-to-worst ordering. To control for position bias, the order of the responses is shuffled independently for each prompt, so no model occupies a fixed slot. We provide the judge prompt used for ranking evaluation for reference in Appendix A.8.

3 Results

3.1 Overall Performance: Diversity vs Quality

Diversity vs Quality Comparison Figure 1 plots average quality against average diversity (DCScore) at the 5,000-sample scale, averaged over all eight datasets. A strong synthetic-data

generator should occupy the upper-right region showcasing it is both diverse *and* high-quality. The Invent API is the only generator that does so, standing well clear of every external system evaluated on both axes simultaneously. It achieves the highest diversity by a wide margin, roughly 19% above the next-best model (GLM-5.3), 24% above Claude Opus 5, 29% above GPT-5.6 Sol, 32% above Gemini-3.1-pro, and 55% above DeepSeek-V4-Pro. At the same time it also achieves the highest quality score (≈ 7.9), about 17% above the strongest baselines (Claude Opus 5 and GLM-5.3, both ≈ 6.75). Appendix A.7 reports the diversity–quality trade-off at all four dataset scales.

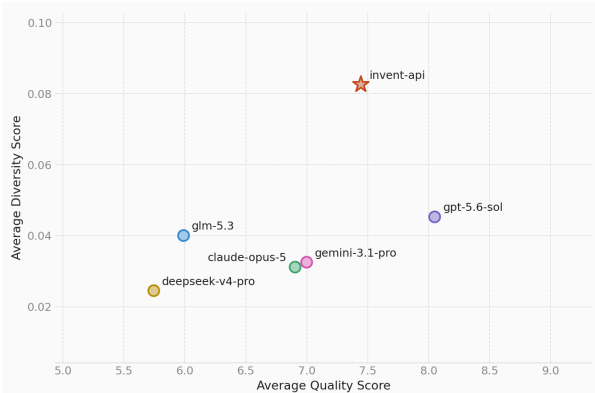
Crucially, the Invent API’s diversity advantage does *not* come at a quality cost, it is pareto-frontier, improving on both objectives at once. Averaged over all datasets at the 5,000-sample scale (Figure 6a), it is the highest-quality generator at 7.90, ahead of Claude Opus 5 (6.76) and GLM-5.3 (6.73) by $\sim 17\%$, and well above Gemini-3.1-pro (5.26) and GPT-5.6 Sol (5.13).

Diversity and Quality across different languages INVENT-A-DATASET supports 244 different languages (Hooker & Roy, 2026), hence it of interest to understand performance across different languages. In Figure 3 we compare Serbian and Hindi (Figure 3a), and show the Invent API is clearly the most diverse (≈ 0.082 , roughly double the next-best baseline) while retaining high-quality generations; GPT-5.6 Sol reaches slightly higher quality (≈ 8.05) but at less than half the diversity, leaving the Invent API alone in the upper-right. Across six English datasets (Figure 3b), the separation is cleaner. The Invent API leads on *both* axes, with the highest quality (≈ 7.5) and the highest diversity (≈ 0.149 , ahead of GLM-5.3 at 0.135). The multilingual setting therefore reinforces the English results, the Invent API’s diversity advantage is not language-specific. Comparing the Invent API to itself, quality is nearly unchanged across languages (≈ 7.5 in English vs. ≈ 7.4 in Serbian and Hindi), while absolute diversity is lower (≈ 0.149 vs. ≈ 0.082); however, every baseline loses more diversity (roughly 60–75% vs. $\approx 44\%$), so the Invent API’s relative advantage widens outside English.

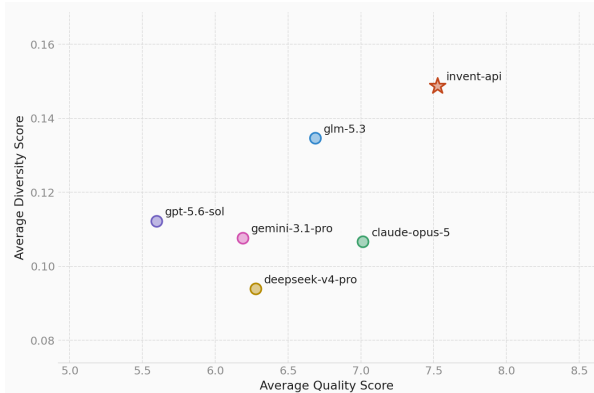
We also measure whether the outputs are generated in the requested language; the results for language confusion (Marchisio et al., 2024) in Hindi are reported in the Appendix A.4. To verify the language of the generated datasets, we applied `langid` (Lui & Baldwin, 2012) to each generated example. All evaluated baselines exhibit minimal language drift, producing the large majority of examples in the target language; the highest drift rate is 7.25%, observed for DeepSeek-V4-Pro. INVENT-A-DATASET yields the lowest drift of all methods, with only 0.15% (less than 1%) of its Hindi outputs identified as a different language.

3.2 Impact of Constraints on Dataset Diversity

Real-world data requests often impose explicit constraints (length, format, structure, persona), which narrow the space of valid outputs and typically reduce diversity. These constraints typically demand more complex generation patterns which increase difficulty of the synthetic data generation task. Figure 4 compares diversity of **unconstrained** dataset queries versus **constrained** (Table 3 and Table 8). Adding constraints reduces diversity for every external baseline, with drops of up to 22%. GPT-5.6 Sol drops 22% ($0.0328 \rightarrow 0.0257$), DeepSeek-v4-Pro 18% ($0.0264 \rightarrow 0.0217$), Gemini-3.1-pro 17% ($0.0306 \rightarrow 0.0254$), Claude Opus 5 15% ($0.0248 \rightarrow 0.0210$) and GLM-5.3 15% ($0.0317 \rightarrow 0.0271$). The Invent API is the only pipeline whose diversity is unaffected (0.0428 in both settings), and no baseline comes close to this robustness. As a result, *under constraints* the Invent API has the highest diversity of any pipeline (0.0428), about 58% above the next best (GLM-5.3,



(a) Serbian and Hindi



(b) English only

Figure 3: Diversity vs. Quality in Multilingual Setting. Diversity–quality trade-off at 5,000 samples for non-English datasets in the constrained setting. In both settings the Invent API occupies the upper-right region, leading on diversity throughout and on quality for the six additional languages.

0.0271). Generating data that fulfills precise constraints is critical for real world modelling and it shows that the Invent API maintains diversity precisely where constraints would otherwise reduce it.

3.3 Diversity and Quality with Scale

Diversity degrades with scale. We also measure how diversity evolves with dataset size. In Figure 5b two effects are noticeable. First, absolute per-sample diversity declines for *all* external generators as the dataset grows: this is expected, since each new sample must differ from an ever-larger pool already generated, so a single model prompted the same way increasingly falls back on repeated patterns and the marginal diversity gain per sample steadily shrinks. Second, and more importantly, the *relative* advantage of the Invent API **widens** with scale. At 2000 samples, Invent API already leads the strongest baseline (glm-5.3) by 0.037 in average diversity (0.250 vs. 0.213, a 17% relative gain), but by 20000 samples this margin grows to 0.072 (0.268 vs. 0.196), a **37%** relative gain that more than doubles the advantage. This widening comes from diverging trends: Invent API’s diversity increases with dataset size, whereas every baseline either plateaus (deepseek-v4-pro, gemini-3.1-pro) or declines (glm-5.3, gpt-5.6-sol, claude-opus-5), suggesting that frontier LLMs increasingly repeat themselves as more data is generated.

Quality with scale. Invent API’s quality lead is stable across scale (Figure 5a). Against GLM-5.3, Invent API leads at every size (8.43 vs. 7.03 at 200, 8.09 vs. 6.76 at 2,000, 7.90 vs. 6.73 at 5,000, and 8.15 vs. 6.55 at 20,000), a relative advantage of roughly 17–24% throughout. Invent API’s own score stays within a narrow band (7.90–8.43) as generation scales, while GLM-5.3 declines steadily (7.03 → 6.55), indicating that Invent API’s quality holds up even as the dataset grows and diversity pressure increases. Taken together with the diversity results, this shows both diversity and quality persist at scale.

Lexical diversity with scale. As mentioned in section 2.2, we complement DCScore based on semantic content with three additional diversity metrics based on textual form. Table 6 reports

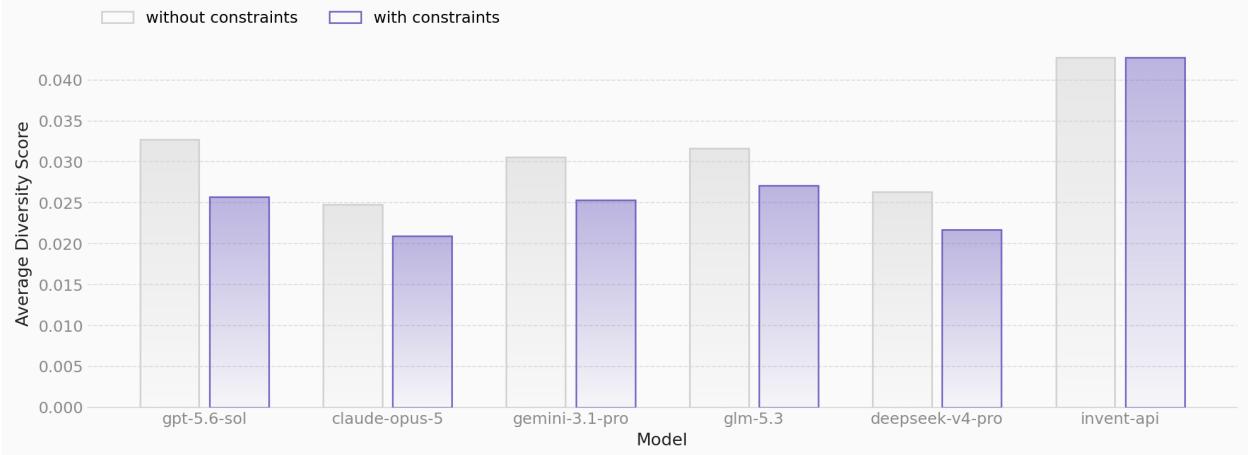


Figure 4: **Diversity under constrained vs. unconstrained settings per generator**, measured with DCScore ($\tau = 0.1$) and averaged across 8 datasets of 20000 samples each. The Invent API is the most robust: its diversity is essentially unchanged (0.0428 in both settings), while every baseline drops by 15–22%. As a result, *under constraints* the Invent API has the highest diversity of any pipeline (0.0428 vs. 0.0271 for the next best, GLM-5.3).

them in the *unconstrained* setting at the 2,000, 5,000, and 20,000 dataset sizes, each computed on a fixed 2,000 prompt random sample so the numbers are directly comparable across sizes. The Invent API is the best generator on *every* metric at *every* scale. Its n -gram diversity is the highest and *increases* with dataset size ($1.75 \rightarrow 1.82 \rightarrow 1.87$). It’s relative gain compared to next closest baseline (GLM-5.3) are 9.4%, 7.1%, and 10.7% at 2K, 5K, and 20K respectively. It also yields the lowest compression ratio at every size (≈ 3.5 , i.e. it has the least repetitive prompts, versus 3.95–5.5 for the baselines). It also has a zero exact-duplicate rate (0.0%) at all scales, while every external baseline outputs 6.2–19.2% *exact-duplicate* prompts, rising with dataset size and worst for DeepSeek-V4-Pro. These metrics highlight that Invent API’s prompts are the most varied, the least compressible, and not repetitive, while frontier LLMs increasingly fall back on repetition as more data is generated. Results for the *constrained* setting are reported in Appendix A.5.

3.4 Relevance of Generated Samples

Relevance of Generated Samples Relevance measures whether a sample matches the dataset specification, independent of how well-constructed or diverse it is. Figure 6b reports average relevance per model at 5,000 samples. While relevance is high across all generators (7.1–8.9), confirming that every generator produces on-specification data, the Invent API sits in the middle rather than at the top. Claude Opus 5 (8.86) and GLM-5.3 (8.66) outperform Invent API by more than a point. The Invent API (7.47) is comparable to Gemini-3.1-pro (7.51), DeepSeek-v4-pro (7.44) and above GPT-5.6 Sol (7.11). Invent API’s current advantage lies in maintaining diversity and quality but currently it does not always produce the most on-specification relevant samples. Improving relevance without sacrificing diversity remains an important future improvement area for the Invent API.

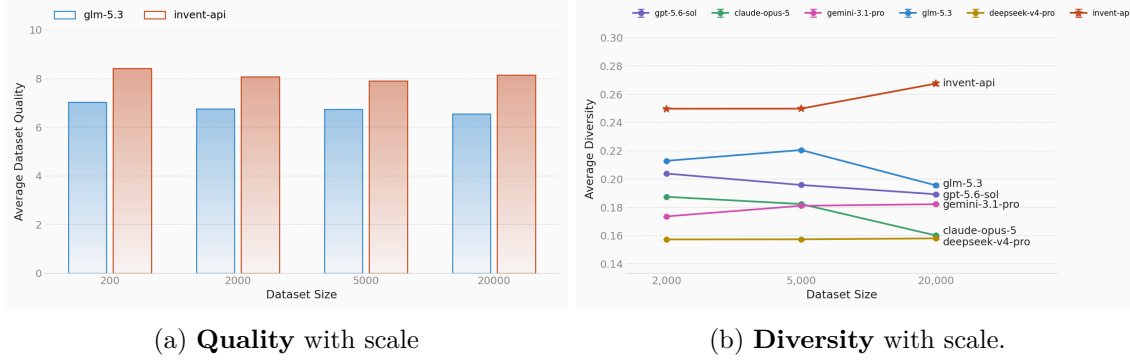


Figure 5: **Left:** Quality with scale. Average dataset quality for GLM-5.3 and the Invent API at 200, 2,000, and 5,000 and 20,000 samples. The Invent API leads at every scale and remains stable as the dataset grows. **Right:** Diversity with scale. Average diversity (DCScore, $\tau = 0.1$) for all generators at 2,000, and 5,000 and 20,000 samples measured on 2,000 samples randomly drawn from each dataset. Per-sample diversity declines with scale for most baseline generators, whereas Invent API’s diversity remains high (and even increases), so its relative advantage grows with scale.

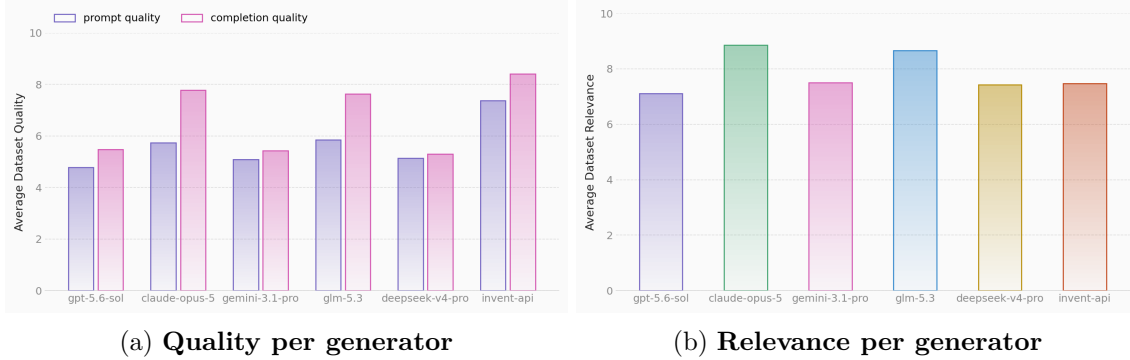


Figure 6: **Left:** Average dataset prompt and completion quality at the 5,000-sample scale. The Invent API is the highest-quality dataset generator, ahead of Claude Opus 5 and GLM-5.3. **Right:** Relevance per generator. Average sample relevance at the 5,000-sample scale. All pipelines produce on-specification data (7.1–8.9); Claude Opus 5 and GLM-5.3 lead, with the Invent API competitive.

3.5 Impact on downstream post-training quality

To test whether better generated data produces a better model, we repeat the following protocol for three models: Llama-3.3-70B, Gemma-4-31B-it and Qwen3.5-9B. For each, we compare the un-tuned base model against five independently fine-tuned variants, each trained on a 20K **Medical QA** dataset produced using a different generator. All six models answer the same set of held-out medical prompts. We use an LLM judge to rank all six responses in a single call from best to worst.

Across all models we finetune, we observe that models trained on Invent API data outperform by a large margin. As seen in Figure 7 (left) Llama-3.3-70B the model trained on Invent API data is ranked first on 54% of the samples. This is more than double the base un-tuned model (25%) and six times the strongest competing fine-tuned model trained on Claude Opus 5 generated data (9%). These patterns hold for Gemma-4-31B-it (Figure 7, right), the four models fine-tuned on data from other generators all fail to beat the base for Top-1 rank share (DeepSeek-V4-Pro (12%), Claude

Data Variant	NGD _{prompt} ↑			CR _{prompt} ↓			Dup _{prompt} (%) ↓		
	2K	5K	20K	2K	5K	20K	2K	5K	20K
Invent API	1.75	1.82	1.87	3.67	3.51	3.47	0.0	0.0	0.0
GLM-5.3	1.57	1.70	1.69	4.70	3.95	3.96	6.2	6.8	8.9
Gemini-3.1-pro	1.53	1.62	1.66	4.59	4.22	4.08	7.5	6.8	6.5
GPT-5.6 Sol	1.60	1.65	1.59	4.33	4.33	4.39	9.6	10.4	12.4
Claude Opus 5	1.55	1.54	1.48	4.27	4.25	4.28	8.5	7.9	8.4
DeepSeek-V4-Pro	1.07	1.07	1.07	5.50	5.53	5.51	19.2	18.5	18.8

Table 6: Lexical prompt diversity and duplication in the *unconstrained* setting, averaged over the eight datasets. Each dataset size (2K/5K/20K) is measured on a fixed 2,000 prompt random sample, so columns are directly comparable. n-gram diversity (NGD): ↑ implies more diverse. Compression Ratio (CR): ↓ implies less redundant; Dup is the percentage of exact-duplicate prompts (↓ is better). Best data variant is indicated in bold for each column.

Opus 5 (10%), GLM-5.3(8%) and GPT-5.6 Sol (8%)). For Qwen3.5-9B, the untuned base is a really strong starting point and edges the Invent API fine-tune on Top-1 rank share (44% vs. 38%), yet Invent API still ranks above every other generator’s fine-tune (see Appendix A.6 for details). These results collectively confirm that differences in data quality and diversity carry through to the models trained on that data. The ordering is consistent with the dataset-level results in Sections 3.1 and 3.3.

4 Related Work

Historically, significant effort has been invested in collecting, labeling, filtering, and reshaping data to fit a training objective (Wang et al., 2024a). A large body of work has emerged which shows that efforts to better curate training corpus, including de-duping (Taylor et al., 2022; Kocetkov et al.), data pruning (Marion et al.; Singh et al., 2024b; Sorscher et al., 2022; Albalak et al., 2024; Tirumala et al., 2023; Chimoto et al., 2024) or data prioritization (Boubdir et al., 2023a; Thakkar et al., 2023) can lead to better model quality. However, most of these works start from a existing data pool and aim to modify it to arrive at higher quality data. The result is often a dataset that only approximates the desired behavior, and a model trained around the limitations of the available data rather than the target capability. In contrast, our work is focused explicitly on the zero data regime where no data is available and optimizing directly for any set of constraints or capabilities a practitioner desires.

Dataset search approaches Dataset search poses unique challenges distinct from traditional web search (Brickley et al., 2019). Users span a wide range of expertise and goals, and in many cases those goals are far removed from the datasets themselves. The datasets are also often difficult to peruse manually. Despite advances in interpreting natural-language intents (Chapman et al., 2020a), users still struggle with incomplete and inconsistent metadata, with expressing information-seeking needs as structured search constraints, and with assessing dataset relevance (Kacprzak et al., 2019; Li et al., 2025b). Crucially, dataset search is fundamentally limited to *retrieving* data that already exists; when no suitable dataset is available, search offers no recourse.

Synthetic data Synthetic data offers several compelling advantages. It allows practitioners to

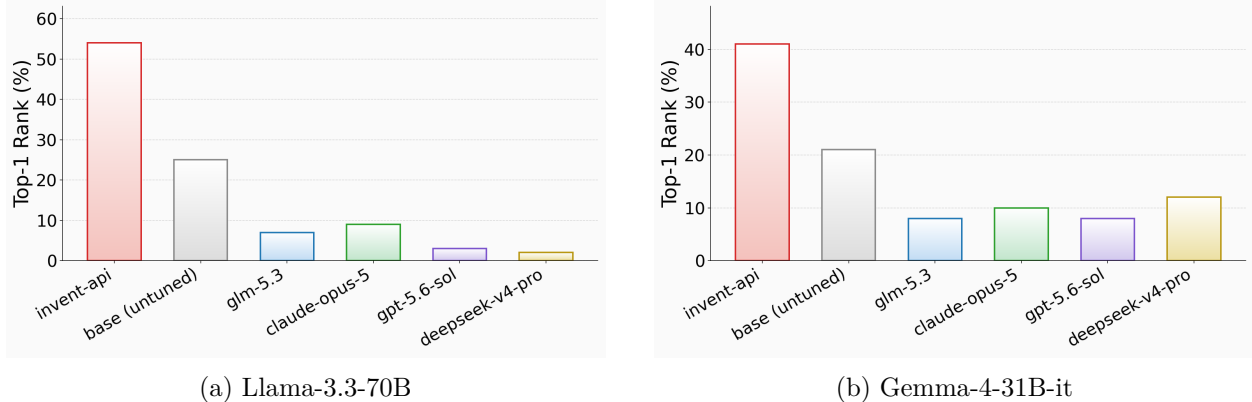


Figure 7: **Top-1 rank share in downstream ranking evaluation.** Percentage of held-out medical test prompts on which each fine-tune is ranked first, with all six responses to a prompt judged together in a single call. Within each panel, the x-axis lists models fine-tuned on different synthetic data variants, plus the untuned base model (grey) as a control. **Left:** Llama-3.3-70B. The Invent API fine-tune is the clear winner, ranking first on 54% of prompts, more than double the untuned base at 25%. **Right:** Gemma-4-31B-it. The Invent API fine-tune again leads at 41%, well above the base at 21%. In both cases, the models fine-tuned on data from other generators (GLM-5.3, Claude Opus 5, GPT-5.6 Sol, DeepSeek-V4-Pro) trail the base by a wide margin.

target long-tail scenarios that are rare in the real distribution (Wang et al., 2024a; Long et al., 2024), minimizes the cost of additional data collection, and can self-recursively increase the difficulty of examples to elicit more advanced capabilities (Xu et al., 2024; Luo et al., 2023). The cost of generating new data has decreased over time, allowing for “on-the-fly” optimization in the data space. (Shimabucoro et al., 2024). These benefits, however, are contingent on the diversity of the generated data. Training on narrow or repetitive synthetic samples degrades performance and, in the recursive limit, induces *model collapse*, in which successive models progressively lose the tails of the distribution (Shumailov et al., 2024; Schaffelder & Gatt, 2026). Maintaining diversity is therefore a central objective for any synthetic data generation method. Most data synthesis approaches leverage existing data structures while adapting them to new requirements. Seed-based methods such as Self-Instruct (Wang et al., 2023b) and Alpaca (Taori et al., 2023) bootstrap large instruction-tuning sets from a small pool of human-written seed examples, while evolutionary methods such as Evol-Instruct/WizardLM (Xu et al., 2024) and its code variant WizardCoder (Luo et al., 2023) iteratively rewrite an existing dataset into more complex and diverse instances. Other work tailors synthetic data to a target model or task, but still conditions on seed instructions or an existing corpus (Wang et al., 2024b). In contrast, our work assumes the end user has *no* access to any existing data. We construct a dataset zero-shot, from a natural-language description of the desired task alone.

5 Conclusion

A systematic weakness of using external model APIs directly is that their generated samples fall into repetitive patterns, which erodes the diversity of the resulting dataset. This effect is clearest when tracking diversity across dataset sizes. When a single model is prompted repeatedly, novel outputs become increasingly hard to produce and per-sample diversity steadily declines. Invent API

is far more resistant to this collapse, and its advantage is evident as we scale across dataset size. At scale, it generates data that is substantially more diverse than every baseline while maintaining the highest overall dataset quality and remaining on-specification.

Limitations

There are some important shortcomings of INVENT-A-DATASET. Today it produces instruction and preference datasets for SFT and alignment training but does not yet extend to tool-call traces or multi-step, agentic trajectories a model produces when driving a harness. We leave this as the subject of future work as harness datasets tend to be more particular to a given environment: the available tools, their signatures and the harness’s control flow. That coupling makes such datasets narrower and faster to go stale as an environment evolves, and harder to reuse across settings than regular SFT and preference datasets supported at present. There is also a valid question of whether this data is best internalized in parametric weight updates. Additionally, INVENT-A-DATASET is limited to text modality at present. In the future, we hope to extend this to support multimodal dataset generation.

References

- Adaption Labs. Invent a dataset. <https://docs.adaptionlabs.ai/adaptive-data/invent-a-dataset/>, 2026. Adaption API documentation. Accessed: 2026-09-24.
- Adaption Research Staff. AutoScientist: Automating the science of model training. Adaption Labs Blog, May 2026. URL <https://adaptionlabs.ai/blog/autoscientist>. Accessed: 2026-09-29.
- Alon Albalak, Yanai Elazar, Sang Michael Xie, Shayne Longpre, Nathan Lambert, Xinyi Wang, Niklas Muennighoff, Bairu Hou, Liangming Pan, Haewon Jeong, et al. A survey on data selection for language models. *arXiv preprint arXiv:2402.16827*, 2024.
- Anthropic. System card: Claude Opus 5. <https://www-cdn.anthropic.com/c5fbac3f0b1280a933ebd26d3cb8bb9f5bdeaf48/Claude%20opus%205%20System%20Card.pdf>, July 2026. Accessed: 2026-09-24.
- Quentin Bertrand, Avishek Joey Bose, Alexandre Duplessis, Marco Jiralerspong, and Gauthier Gidel. On the stability of iterative retraining of generative models on their own data. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.00429.
- Meriem Boubdir, Edward Kim, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. Which prompts make the difference? data prioritization for efficient human llm evaluation. *arXiv preprint arXiv:2310.14424*, 2023a.
- Meriem Boubdir, Edward Kim, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. Which prompts make the difference? data prioritization for efficient human llm evaluation, 2023b. URL <https://arxiv.org/abs/2310.14424>.
- Dan Brickley, Matthew Burgess, and Natasha Noy. Google dataset search: Building a search engine for datasets in an open web ecosystem. In *The World Wide Web Conference (WWW)*, pp. 1365–1375, 2019. doi: 10.1145/3308558.3313685.

-
- Martin Briesch, Dominik Sobania, and Franz Rothlauf. Large language models suffer from their own output: An analysis of the self-consuming training loop. *arXiv preprint arXiv:2311.16822*, 2023.
- Adriane Chapman, Elena Simperl, Laura Koesten, George Konstantinidis, Luis-Daniel Ibáñez, Emilia Kacprzak, and Paul Groth. Dataset search: A survey. *The VLDB Journal*, 29:251–272, 2020a.
- Adriane Chapman, Elena Simperl, Laura Koesten, George Konstantinidis, Luis-Daniel Ibáñez, Emilia Kacprzak, and Paul Groth. Dataset search: a survey. *The VLDB Journal*, 29(1):251–272, 2020b.
- Everlyn Asiko Chimoto, Jay Gala, Orevaoghene Ahia, Julia Kreutzer, Bruce A Bassett, and Sara Hooker. Critical learning periods: Leveraging early training dynamics for efficient data pruning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 9407–9426, 2024.
- DeepSeek-AI. DeepSeek-V4-Pro-0813. <https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro-0813>, August 2026. Model card. Accessed: 2026-09-24.
- Elvis Dohmatob, Yunzhen Feng, and Julia Kempe. Model collapse demystified: The case of regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2402.07712.
- Google DeepMind. Gemini 3.1 Pro – model card. <https://deepmind.google/models/model-cards/gemini-3-1-pro/>, February 2026. Accessed: 2026-09-24.
- Yanzhu Guo, Guokan Shang, Michalis Vazirgiannis, and Chloé Clavel. The curious decline of linguistic diversity: Training language models on synthetic text. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 3589–3604. Association for Computational Linguistics, 2024. URL <https://aclanthology.org/2024.findings-naacl.228/>.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. *arXiv preprint arXiv:1904.09751*, 2019.
- Sara Hooker and Sudip Roy. Expand your world. Adaption Labs Blog, April 2026. URL <https://adaptionlabs.ai/blog/expand-your-world>. Accessed: 2026-09-30.
- Yuxin Jiang, Yufei Wang, Xingshan Zeng, Wanjun Zhong, Liangyou Li, Fei Mi, Lifeng Shang, Xin Jiang, Qun Liu, and Wei Wang. Followbench: A multi-level fine-grained constraints following benchmark for large language models, 2024. URL <https://arxiv.org/abs/2310.20410>.
- Emilia Kacprzak, Laura Koesten, Luis-Daniel Ibáñez, Tom Blount, Jeni Tennison, and Elena Simperl. Characterising dataset search queries. *Companion of The Web Conference*, 2019.
- Denis Kocetkov, Raymond Li, Loubna Ben Allal, Jia Li, Chenghao Mou, Carlos Muñoz Ferrandis, Yacine Jernite, Margaret Mitchell, Sean Hughes, Thomas Wolf, et al. The stack: 3 tb of permissively licensed source code, 2022. URL <https://arxiv.org/abs/2211.15533>.
- Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Bal-subramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton Earnshaw, Imran Haque, Sara M Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. Wilds: A benchmark of in-the-wild distribution shifts. In Marina Meila and Tong Zhang

-
- (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 5637–5664. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/koh21a.html>.
- Ilia Kulikov, Chenxi Whitehouse, Tianhao Wu, Yixin Nie, Swarnadeep Saha, Eryk Helenowski, Weizhe Yuan, Olga Golovneva, Jack Lanchantin, Yoram Bachrach, Jakob Foerster, Xian Li, Han Fang, Sainbayar Sukhbaatar, and Jason Weston. Autodata: An agentic data scientist to create high quality synthetic data, 2026. URL <https://arxiv.org/abs/2606.25996>.
- Keyu Li, Mohan Jiang, Dayuan Fu, Yunze Wu, Xiangkun Hu, Dequan Wang, and Pengfei Liu. Datasetresearch: Benchmarking agent systems for demand-driven dataset discovery. *arXiv preprint arXiv:2508.06960*, 2025a.
- Pengyue Li, Sheng Wang, Hua Dai, Zhiyu Chen, Zhifeng Bao, and Brian D. Davison. A survey on open dataset search in the llm era: Retrospectives and perspectives, 2025b. URL <https://arxiv.org/abs/2509.00728>.
- Rachel Lin, Bhavya Chopra, Wenjing Lin, Shreya Shankar, Madelon Hulsebos, and Aditya G Parameswaran. Rethinking dataset discovery with datascout. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–16, 2025.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: Nlg evaluation using gpt-4 with better human alignment, 2023. URL <https://arxiv.org/abs/2303.16634>.
- Lin Long, Rui Wang, Ruixuan Xiao, Junbo Zhao, Xiao Ding, Gang Chen, and Haobo Wang. On LLMs-driven synthetic data generation, curation, and evaluation: A survey. *arXiv preprint arXiv:2406.15126*, 2024.
- Shayne Longpre, Robert Mahari, Anthony Chen, Naana Obeng-Marnu, Damien Sileo, William Brannon, Niklas Muennighoff, Nathan Khazam, Jad Kabbara, Kartik Perisetla, et al. A large-scale audit of dataset licensing and attribution in ai. *Nature Machine Intelligence*, 6(8):975–987, 2024.
- Marco Lui and Timothy Baldwin. langid. py: An off-the-shelf language identification tool. In *Proceedings of the ACL 2012 system demonstrations*, pp. 25–30, 2012.
- Ziyang Luo, Can Xu, Pu Zhao, Qingfeng Sun, Xiubo Geng, Wenxiang Hu, Chongyang Tao, Jing Ma, Qingwei Lin, and Daxin Jiang. WizardCoder: Empowering code large language models with evol-instruct. *arXiv preprint arXiv:2306.08568*, 2023.
- Kelly Marchisio, Wei-Yin Ko, Alexandre Berard, Théo Dehaze, and Sebastian Ruder. Understanding and mitigating language confusion in LLMs. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 6653–6677, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.380. URL <https://aclanthology.org/2024.emnlp-main.380/>.
- Max Marion, Ahmet Üstün, Luiza Pozzobon, Alex Wang, Marzieh Fadaee, and Sara Hooker. When less is more: Investigating data pruning for pretraining llms at scale, 2023. URL <https://arxiv.org/abs/2309.04564>.

-
- OpenAI. GPT-5.6 preview system card. <https://deploymentsafety.openai.com/gpt-5-6-preview>, June 2026. Accessed: 2026-09-24.
- OpenRouter. OpenRouter: A unified interface for LLMs. <https://openrouter.ai>, 2026. Accessed: 2026-09-24.
- Vishakh Padmakumar and He He. Does writing with language models reduce content diversity?, 2024. URL <https://arxiv.org/abs/2309.05196>.
- Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In Kamalika Chaudhuri and Ruslan Salakhutdinov (eds.), *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 5389–5400. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/recht19a.html>.
- Yuji Roh, Geon Heo, and Steven Euijong Whang. A survey on data collection for machine learning: a big data-ai integration perspective. *IEEE Transactions on Knowledge and Data Engineering*, 33(4):1328–1347, 2019.
- Israfel Salazar, Manuel Fernández Burda, Shayekh Islam, Arshia Soltani Moakhar, Shivalika Singh, Fabian Farestam, Angelika Romanou, Danylo Boiko, Dipika Khullar, Mike Zhang, et al. Kaleidoscope: In-language exams for massively multilingual vision evaluation. In *International Conference on Learning Representations*, volume 2026, pp. 81112–81164, 2026.
- Max Schaffelder and Albert Gatt. Synthetic eggs in many baskets: The impact of synthetic data diversity on LLM fine-tuning. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens (eds.), *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 7265–7293, San Diego, California, United States, July 2026. Association for Computational Linguistics. ISBN 979-8-89176-395-1. doi: 10.18653/v1/2026.findings-acl.360. URL <https://aclanthology.org/2026.findings-acl.360/>.
- Chantal Shaib, Venkata S. Govindarajan, Joe Barrow, Jiuding Sun, Alexa F. Siu, Byron C. Wallace, and Ani Nenkova. Standardizing the measurement of text diversity: A tool and a comparative analysis of scores, 2026. URL <https://arxiv.org/abs/2403.00553>.
- Luísa Shimabucoro, Sebastian Ruder, Julia Kreutzer, Marzieh Fadaee, and Sara Hooker. LLM see, LLM do: Leveraging active inheritance to target non-differentiable objectives. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 9243–9267, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.521. URL <https://aclanthology.org/2024.emnlp-main.521/>.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget. *arXiv preprint arXiv:2305.17493*, 2023.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. Ai models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024. doi: 10.1038/s41586-024-07566-y.
- Shivalika Singh, Freddie Vargus, Daniel D’souza, Börje Karlsson, Abinaya Mahendiran, Wei-Yin Ko, Herumb Shandilya, Jay Patel, Deividas Mataciunas, Laura O’Mahony, Mike Zhang, Ramith

-
- Hettiarachchi, Joseph Wilson, Marina Machado, Luisa Moura, Dominik Krzemiński, Hakimeh Fadaei, Irem Ergun, Ifeoma Okoh, Aisha Alaagib, Oshan Mudannayake, Zaid Alyafeai, Vu Chien, Sebastian Ruder, Surya Guthikonda, Emad Alghamdi, Sebastian Gehrmann, Niklas Muennighoff, Max Bartolo, Julia Kreutzer, Ahmet Üstün, Marzieh Fadaee, and Sara Hooker. Aya dataset: An open-access collection for multilingual instruction tuning. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11521–11567, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.620. URL <https://aclanthology.org/2024.acl-long.620/>.
- Shivalika Singh, Freddie Vargus, Daniel D’souza, Börje F Karlsson, Abinaya Mahendiran, Wei-Yin Ko, Herumb Shandilya, Jay Patel, Deividas Mataciunas, Laura O’Mahony, et al. Aya dataset: An open-access collection for multilingual instruction tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11521–11567, 2024b.
- Shivalika Singh, Angelika Romanou, Clémentine Fourrier, David Ifeoluwa Adelani, Jian Gang Ngui, Daniel Vila-Suero, Peerat Limkonchotiwat, Kelly Marchisio, Wei Qi Leong, Yosephine Susanto, et al. Global mmlu: Understanding and addressing cultural and linguistic biases in multilingual evaluation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 18761–18799, 2025.
- Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari Morcos. Beyond neural scaling laws: beating power law scaling via data pruning. *Advances in Neural Information Processing Systems*, 35:19523–19536, 2022.
- Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023.
- Ross Taylor, Marcin Kardas, Guillem Cucurull, Thomas Scialom, Anthony Hartshorn, Elvis Saravia, Andrew Poulton, Viktor Kerkez, and Robert Stojnic. Galactica: A large language model for science. *arXiv preprint arXiv:2211.09085*, 2022.
- Damien Teney, Ehsan Abbasnejad, Kushal Kafle, Robik Shrestha, Christopher Kanan, and Anton van den Hengel. On the value of out-of-distribution testing: An example of goodhart’s law. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 407–417. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/045117b0e0a11a242b9765e79cbf113f-Paper.pdf.
- Megh Thakkar, Tolga Bolukbasi, Sriram Ganapathy, Shikhar Vashishth, Sarath Chandar, and Partha Talukdar. Self-influence guided data reweighting for language model pre-training. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 2033–2045, 2023.
- Kushal Tirumala, Daniel Simig, Armen Aghajanyan, and Ari Morcos. D4: Improving llm pretraining via document de-duplication and diversification. *Advances in Neural Information Processing Systems*, 36:53983–53995, 2023.

-
- A. Torralba and A. A. Efros. Unbiased look at dataset bias. In *Proceedings of the 2011 IEEE Conference on Computer Vision and Pattern Recognition*, CVPR '11, pp. 1521–1528, USA, 2011. IEEE Computer Society. ISBN 9781457703942. doi: 10.1109/CVPR.2011.5995347. URL <https://doi.org/10.1109/CVPR.2011.5995347>.
- Ke Wang et al. A survey on data synthesis and augmentation for large language models. *arXiv preprint arXiv:2410.12896*, 2024a.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, pp. 13484–13508, 2023a.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-Instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023b. arXiv:2212.10560.
- Zifeng Wang, Chun-Liang Li, Vincent Perot, Long T. Le, Jin Miao, Zizhao Zhang, Chen-Yu Lee, and Tomas Pfister. CodecLM: Aligning language models with tailored synthetic data. In *Findings of NAACL*, 2024b. arXiv:2404.05875.
- Bosi Wen, Pei Ke, Xiaotao Gu, Lindong Wu, Hao Huang, Jinfeng Zhou, Wenchuang Li, Binxin Hu, Wendy Gao, Jiaxin Xu, Yiming Liu, Jie Tang, Hongning Wang, and Minlie Huang. Benchmarking complex instruction-following with multiple constraints composition, 2024. URL <https://arxiv.org/abs/2407.03978>.
- Kyle Wiggers. ‘open’ AI model licenses often carry concerning restrictions. <https://techcrunch.com/2025/03/14/open-model-licenses-often-carry-concerning-restrictions/>, March 2025. TechCrunch. Accessed: 2026-09-24.
- Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, Qingwei Lin, and Daxin Jiang. Wizardlm: Empowering large pre-trained language models to follow complex instructions. In *International Conference on Learning Representations*, volume 2024, pp. 30745–30766, 2024.
- Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, Huayang Li, and Jian Li. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation, 2022. URL <https://arxiv.org/abs/2206.02369>.
- Z.ai. GLM-5.3. <https://huggingface.co/zai-org/GLM-5.3>, August 2026. Model card. Accessed: 2026-09-24.
- Biao Zhang, Zhongtao Liu, Colin Cherry, and Orhan Firat. When scaling meets llm finetuning: The effect of data, model and finetuning method, 2024. URL <https://arxiv.org/abs/2402.17193>.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, et al. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023. URL <https://arxiv.org/abs/2311.07911>.

Yuchang Zhu, Huizhe Zhang, Bingzhe Wu, Jintang Li, Zibin Zheng, Peilin Zhao, Liang Chen, and Yatao Bian. Measuring diversity in synthetic datasets. *arXiv preprint arXiv:2502.08512*, 2025.

A Appendix

A.1 Quality Score Description

Prompt quality. This rubric grades the generated prompt on a 0–10 scale based on criteria such as specificity, scope, and completeness. Low scores (1–4) denote incoherent, vague, or under-specified requests; mid scores (5–6) a clear, executable task lacking depth or context; and high scores (7–10) well-structured prompts with explicit success criteria. Crucially, prompts are judged *within* the form the dataset request specifies, so a deliberately short prompt is not penalized for brevity it was asked to have by the user.

Completion Quality: grades the generated response based on a 0–10 based on its overall quality, structure, and adherence to the prompt’s stated constraints. Low scores (1–2) denote incoherent, incorrect, or incomplete answers; mid scores (3–5) a competent, accurate response covering the main points but lacking depth; and high scores (6–10) comprehensive, well-organized answers that address caveats and edge cases. As with prompt quality, depth is judged relative to any length, format, or style limits the dataset request imposes.

Below we show full judge prompt templates:

Quality and Relevance Judge Template

A strict, meticulous judge for curating supervised fine-tuning datasets. Given a **target specification** (a dataset-level data request plus language tags) and one **candidate** prompt/completion pair, the judge decides whether the candidate belongs in the dataset and how good it is.

- **Scoring axes:** relevance, prompt quality, completion quality — each an integer 0–10.
- **Output:** a single compact JSON object, no other text.
- **Template placeholders:** `{query}`, `{language}`, `{prompt}`, `{completion}`, `{prompt_quality_rubric}`, `{completion_quality_rubric}`.

System Message

You are a strict, meticulous judge who curates supervised fine-tuning datasets. Given a TARGET SPECIFICATION (a dataset-level data request and language tags) and one CANDIDATE prompt/completion pair, you decide whether the candidate genuinely BELONGS in that dataset and how good it is. Be decisive and consistent: reward concrete, self-contained, on-topic prompts paired with accurate, complete completions; penalize off-topic, wrong-language,

incomplete/truncated, answer-shaped, or meta (“generate a dataset of...”) prompts and shallow, wrong, or evasive completions. You score the candidate on THREE INDEPENDENT AXES — relevance (does the sample belong in the dataset?), prompt quality (how good is the prompt on its own?), and completion quality (how good is the completion as an answer to the prompt?) — each 0–10. Reply with ONLY a single compact JSON object and no other text.

Prompt Quality Rubric

Assign a quality score from 0 to 10 for the overall quality of the prompt.

Scale

0 (Empty prompt) — There is no prompt. It is a null entry.

1–2 (Incoherent / Nonsensical) — The request is unreadable, random, or missing meaning. Single words. Gibberish, unrelated fragments, no clear task. May be impossible to answer given the available context. No discernible task, goal, or meaning. Examples: “asdf quantum banana”, “Help”, “AI”, “Write something”, “???”

3–4 (Unclear or Under-Specified) — Readable but lacks sufficient detail to execute effectively. Identifiable topic but missing critical parameters that specify scope, audience, format or constraints. Overly general request. Examples: “Tell me about technology”, “Write about climate change”, “Explain machine learning”, “What do you think?”

5–6 (Basic Clarity) — Identifiable task with clear action verb. Straightforward but lacks depth or strategic context. Examples: “Summarize this article”, “Write a 3-paragraph explanation for beginners”, “List 5 pros and cons of remote work”

7–8 (Well-Structured) — Includes relevant context or use case. Clear success criteria or deliverable specifications. May request comparative analysis or structured reasoning. Example: “Write a 150-word summary of this AI safety paper for a tech newsletter audience, emphasizing practical implications”

9–10 (Expert-Level / Meta-Optimized) — Advanced prompt engineering with sophisticated scaffolding. Explicit role assignment or persona specification. Multi-dimensional reasoning or evaluation criteria. Anticipates edge cases, nuance, or quality standards. May include self-correction mechanisms or iterative instructions.

Special Cases

- **Respect dataset-request constraints:** Do NOT penalize a prompt for lacking detail, length, context, or scaffolding that the dataset request explicitly told it to omit or cap. When the request mandates a form — e.g., short or one-line prompts, no provided context, a fixed template, or a plain/casual register — score how well the prompt is crafted *within* that form: is it clear, purposeful, and complete for its intended scope? A deliberately terse prompt that cleanly specifies its task can still score 7+; reserve 3–4 for prompts that are vague or under-specified *beyond* what the request requires. Judge specification quality relative to the requested form, not against an unconstrained ideal.
- **Multi-turn conversation:** if the prompt is a chat involving a multi-turn conversation, score based on the last turn in the conversation.
- **A complex prompt that is poorly asked:** Focus on prompt structure and specification,

not topic complexity. A difficult question should still be ranked low for quality if it is poorly asked or if the desired output is not clear.

- **Long prompts are not necessarily high quality:** Watch out for verbose but unfocused prompts masquerading as detailed. If a request is needlessly lengthy, or lacks helpful structure, it should be scored as low quality.

Completion Quality Rubric

Assign a quality score from 0 to 10 for the completion AS AN ANSWER TO THE PROMPT.

Scale

0 (Empty completion) — There is no response; it is a null value.

1 (Incoherent / Nonsensical) — Unreadable response, completely wrong answer, or fails to address the prompt. Gibberish, broken text, or semantically empty output. Completely misunderstands or ignores the prompt. Addresses an entirely wrong topic or provides irrelevant information. Contains critical factual errors that make the response unusable. May acknowledge the prompt but provides no actual answer.

2 (Inadequate or Incomplete) — Addresses the prompt but with major gaps or issues. Provides only trivially basic or superficial information. Answers some parts of the prompt while ignoring others. Contains significant accuracy issues or misleading statements. Missing critical context, details, or qualifications. Oversimplified to the point of being unhelpful. Disorganized or poorly structured.

3–5 (Basic Clarity) — Competent response that meets basic requirements. Directly addresses the prompt. Factually accurate with no major errors. Reasonably complete coverage of main points. Clear and understandable structure. May include basic examples or context. Sufficient for basic understanding but lacks depth or nuance.

6–8 (Well-Structured) — Comprehensive coverage with relevant details and context. Deep, nuanced understanding with multiple perspectives. Includes helpful examples, analogies, or illustrations. Well-organized with clear logical flow. Addresses edge cases, caveats, limitations, or trade-offs. Anticipates follow-up questions or potential misconceptions. Provides practical, actionable guidance.

9–10 (Expert-Level / Meta-Optimized) — Goes meaningfully beyond what was asked (in helpful ways). Demonstrates mastery with strategic depth and synthesis. Provides novel insights, non-obvious connections, or transferable frameworks. Identifies unstated assumptions or addresses the underlying need. Structured for maximum clarity and usability (may include tables, schemas, decision trees). Anticipates and prevents common mistakes or misunderstandings. Teaches underlying principles, not just answers the prompt.

Special Cases

- **Respect dataset-request constraints:** Do NOT penalize a completion for lacking depth, length, elaboration, examples, or nuance that the dataset request explicitly excluded or capped. When the request mandates the completion's length, format, style, or depth (e.g., "one-sentence answers", "terse", "bullet list only", "no explanation"), score how fully and accurately it satisfies the prompt *within* those bounds. A concise, correct, well-executed answer that obeys a brevity or format requirement can still score 6+ (or higher if

it executes the constrained form excellently); do not treat requested brevity or simplicity as “inadequate” or “superficial”. Judge depth relative to the requested form, not against an unconstrained ideal.

- **Multi-turn conversation:** score the completion as a response to the last user turn.
- **Long completions are not necessarily high quality:** verbose, padded, or repetitive answers that do not add substance should be scored as low quality.

Complete Judge Prompt

Target specification

- Dataset data request (task & topic; applies to the whole dataset): `{query}`
- Target language: `{language}`

Score the candidate sample containing prompt and completion on THREE INDEPENDENT AXES, each an integer 0–10.

Axis 1 — `relevance_score`

Does the candidate SAMPLE belong in the target dataset? Membership requires FOUR HARD CRITERIA, all of which must hold. They are listed in order of importance — (A) and (B) are foundational, and (C) outranks (D): a clear violation of an explicitly stated constraint is a more serious membership failure than a prompt-usability defect.

- **(A) Task & Topic** — the sample matches the dataset request’s task type *and* subject matter.
- **(B) Language** — the sample is written in the target language the request specifies.
- **(C) Constraint Following** — the sample obeys the constraints the dataset request EXPLICITLY states for the prompt and/or completion (e.g. length, format, structure, style, persona). Only constraints the request actually states count; if the request states no constraints, treat (C) as satisfied — do NOT invent constraints.
- **(D) Usable Prompt** — the prompt is a self-contained, complete, standalone user prompt: NOT an answer/completion, NOT a truncated fragment, NOT a meta-instruction about building or generating a dataset, NOT off-topic filler, and NOT dependent on hallucinated or absent context (e.g. referring to a file, document, image, table, link, or earlier conversation that is not actually included in the prompt).

`relevance_score` scale

0 (Empty) — No prompt or completion / null / unreadable.

1–2 (Does not belong) — Fails (A) or (B): off-topic for the dataset request’s task/subject, wrong task type, or wrong language. E.g. target = customer-support inquiries, candidate = “The mitochondria is the powerhouse of the cell.”

3 (Constraint violation) — On-topic and in-language, but clearly violates an explicitly stated (C) constraint (e.g. wrong required format, outside a stated length, missing a mandated structure). Because (C) outranks (D), score this LOWER than a mere usability defect.

4 (Unusable prompt) — On-topic, in-language, and not in clear violation of a stated constraint, but fails (D): it is an answer/completion rather than a prompt, a bare fragment or gibberish, a truncated/incomplete request, a meta / data-generation instruction (e.g. “Generate 50 billing dispute tickets”), or it relies on absent/hallucinated context (e.g. “summarize the attached file”, “based on the table above”).

5–6 (Borderline member) — Meets (A), (B), (D) and commits no outright (C) violation, but follows the stated constraints only PARTIALLY or loosely, and/or is generic, under-specified, or only tangential to the requested subject.

7–8 (Belongs) — Meets (A), (B), (D) and fully follows every stated constraint (C): clearly matches task + topic + language, satisfies all requested constraints, and is a complete, concrete, standalone prompt with enough specificity to be a solid example. Minor gaps only.

9–10 (Excellent fit) — Meets all four criteria, follows every stated constraint precisely and completely, and is realistic, specific, and fully self-contained — an ideal member of this dataset.

Relevance notes. First read the free-form dataset request to extract (A) task + topic, (B) any language requirement, and (C) any EXPLICIT form/length/format/style/persona constraints. Priority order is (A)/(B) > (C) > (D): wrong language or topic caps at 1–2; a clear violation of a stated constraint (C) caps at ≤ 3 ; a usability defect (D) caps at ≤ 4 . Full compliance with all stated constraints is required to score 7+, and partial/loose compliance caps at 5–6. If the request states no constraints, (C) is satisfied — do not cap the score for it and do not invent constraints. For multi-turn candidates, judge the whole conversation, anchored on the user’s request(s).

Axis 2 — `prompt_quality_score`

How good is the candidate PROMPT on its own, independent of the dataset it targets? Use the **Prompt Quality Rubric** above (`{prompt_quality_rubric}`; score the user prompt only). Prompt quality is INDEPENDENT of relevance: an off-topic prompt can still be well-written (high quality, low relevance), and a perfectly on-topic prompt can be sloppy or under-specified (high relevance, low quality). Score the two axes separately. However, do NOT penalize prompt quality for a property the dataset request explicitly requires. If the request mandates a form — e.g., short/one-line prompts, no supplied context, a fixed template, plain casual phrasing, a specific length or format — judge how well the prompt executes within that form, not whether it has the depth, length, or scaffolding that would score higher for an unconstrained prompt. A prompt that is concise because the request demanded concision is not “under-specified.”

Axis 3 — `completion_quality_score`

How good is the candidate COMPLETION as an answer to the candidate prompt? Use the **Completion Quality Rubric** above (`{completion_quality_rubric}`; score the completion only, in light of what the prompt asks for and the dataset request’s expectations). Completion quality is INDEPENDENT of the other two axes: a great answer to an off-topic or sloppy prompt still scores high here, and a weak answer to an excellent prompt scores low. However, do NOT penalize completion quality for a property the dataset request explicitly requires. If the request mandates the completion’s length, format, style, or depth (e.g., “one-sentence answers,” “terse,” “bullet list only,” “no explanation”), judge how well the completion

fulfills the prompt within those bounds, not whether it has the elaboration, examples, or nuance that would score higher for an unconstrained answer.

Decision

Set **relevant** to “yes” ONLY if (A) AND (B) AND (C) AND (D) all hold — this always corresponds to **relevance_score** ≥ 5 ; otherwise set it to “no” (**relevance_score** ≤ 4). (C) “holds” when the sample commits no clear violation of any explicitly stated constraint; (D) “holds” when the prompt is usable as defined above. Keep **relevance_score** and the yes/no decision consistent.

Return only this JSON object and nothing else

```
{"relevance_score": <int 0-10>, "prompt_quality_score": <int 0-10>,  
"completion_quality_score": <int 0-10>, "relevant": <"yes" or "no">, "reason":  
<string  $\leq$  30 words naming the deciding criteria for relevance, prompt quality  
and completion quality>}
```

CANDIDATE PROMPT: {prompt}

CANDIDATE COMPLETION: {completion}

A.2 Quality metrics per dataset

Figures 8 and 9 report per-dataset quality results at all 4 scales for the constrained and unconstrained settings, respectively.

A.3 Dataset Generation Queries

Table 8 reports the unconstrained and constrained queries used to generate other datasets in our evaluation set.

A.4 Output Language Consistency

Table 10 reports output language consistency on **hindi_news** (2000 samples from a pool of 20000). We count any output not detected as Hindi as a language error, including outputs in Marathi and Nepali. Invent API was the most consistent system, with only 3 non-Hindi outputs (0.15%), ahead of Claude Opus 5 (7; 0.35%), GLM-5.3 (49; 2.45%), GPT-5.6-sol (50; 2.5%), Gemini 3.1 Pro (77; 3.85%), and DeepSeek V4 Pro (145; 7.25%). Marathi was the most common error for every system except GLM-5.3, whose errors were mostly English (38 of 49). DeepSeek V4 Pro’s error rate was driven almost entirely by Marathi (144 outputs). The only other languages observed were Nepali (DeepSeek V4 Pro and Gemini 3.1 Pro, three outputs combined) and a single Lithuanian output from Invent API.

A.5 Lexical diversity: constrained setting

Table 11 reports the same three lexical prompt metrics in the constrained setting, at the 2,000, 5,000, and 20,000-sample scales (each on a fixed 2,000 prompt random sample). The pattern mirrors the unconstrained results in Table 6. The Invent API has the least redundant prompts (lowest compression ratio at every size) and a near-zero duplicate rate at every scale, while baseline duplication grows with dataset size. On raw n -gram diversity the Invent API and GLM-5.3 are close and alternate at the top (GLM-5.3 edges ahead at 5,000). Overall, constraining generation

Dataset	Unconstrained Query	Constrained Query
Customer Support Banking	Dataset of customer service responses guiding users through credit card activation, blocking, and mortgage inquiries.	Dataset of customer service conversations guiding frustrated users through credit card activation, blocking, and mortgage inquiries. The customer service agent is always welcoming and polite.
Hindi News	Dataset containing Hindi news articles sourced from various Indian news websites, paired with their corresponding headlines and summaries.	Dataset containing Hindi news articles sourced from various Indian news websites about sports, paired with their corresponding summaries. The prompt is always the article headline, and the completion is the article summary.
Customer Support General	A dataset of customer support inquiries and corresponding responses, covering issues such as product information, technical troubleshooting, billing disputes, and service returns.	
Medical QA	Dataset of medical question-answer pairs covering diagnoses, treatments, drug explanations, and healthcare roles. Each entry includes a prompt and a detailed, expert-generated response.	
Hotel Reviews	A collection of customer reviews describing hotel stays, focusing on service, room quality, location, and amenities. The dataset should include both positive and negative experiences.	

Table 8: The single request queries used to generate datasets. Queries are chosen to represent a range of 1) task types, 2) languages and 3) domains. We evaluate on both **unconstrained** and **constrained** versions of the query, where the constrained query requires additional requirements such as 1) tone constraints, 2) json format, 3) multiple choice output restrictions.

lowers absolute duplication rate for all baseline generators whereas Invent API saw a minor increase from 0.0% (in unconstrained setting) to 0.1% (in constrained setting).

A.6 Post-training results

Apart from Llama-3.3-70B and Gemma-4-31B-it, we also fine-tune Qwen3.5-9B. As shown in Figure 10, Qwen3.5-9B is a really strong starting point, edging Invent API fine-tuned variant on both first-place rate (44% vs. 38%) and mean rank (2.17 vs. 2.28). Invent API fine-tune nonetheless remains ahead of every variant corresponding to the other generators.

Across all three chosen model architectures, Invent API is consistently the top-ranked generator relative to the fine-tunes from every other generator. We note that these results are obtained with a modest 20K fine-tuning set. Prior work shows that supervised fine-tuning performance scales predictably with the amount of training data (Zhang et al., 2024), so we expect the margin of the Invent API fine-tune over the strongest base models to widen further at larger data budgets.

A.7 Dataset Diversity vs Dataset Quality Across Different Scales

Figures 11–14 plot average diversity against average quality for datasets of 200, 2,000, 5,000, and 20,000 samples: **invent-api** achieves the highest quality at every scale and the highest diversity at every scale except 200 samples, where all APIs are on par.

Model	Not in language	%	Detected languages
claude-opus-5	7/2000	0.35	hi 1993, mr 7
deepseek-v4-pro-0813	145/2000	7.25	hi 1855, mr 144, ne 1
gemini-3.1-pro	77/2000	3.85	hi 1923, mr 75, ne 2
gpt-5.6-sol	50/2000	2.50	hi 1950, mr 50
glm-5.3	49/2000	2.45	hi 1951, en 38, mr 11
invent-api	3/2000	0.15	hi 1997, mr 2, lt 1

Table 10: Output language consistency on `hindi_news` (2k samples drawn from 20k). *Not in language* counts all outputs detected as any language other than Hindi, including Marathi and Nepali. Language codes: hi = Hindi, mr = Marathi, ne = Nepali, en = English, lt = Lithuanian.

Model	NGD _{prompt} ↑			CR _{prompt} ↓			Dup _{prompt} (%) ↓		
	2K	5K	20K	2K	5K	20K	2K	5K	20K
Invent API	1.69	1.69	1.72	3.55	3.53	3.50	0.1	0.1	0.1
GLM-5.3	1.65	1.80	1.71	4.59	3.75	3.83	0.9	1.2	3.9
Gemini-3.1-pro	1.34	1.39	1.53	5.18	4.75	4.03	2.2	2.3	1.8
GPT-5.6 Sol	1.56	1.57	1.53	4.16	4.22	4.29	2.2	3.7	5.0
Claude Opus 5	1.61	1.56	1.46	4.00	4.02	4.09	2.2	3.9	4.3
DeepSeek-V4-Pro	1.12	1.13	1.12	5.15	5.12	5.17	11.5	11.0	11.6

Table 11: Lexical diversity and duplication in the *constrained* setting, averaged over the eight datasets (computed for each dataset size on a fixed 2,000 prompt random sample). n-gram diversity, NGD (↑ is better), Compression Ratio, CR (↓ is better), Duplicate Rate (Dup %, ↓ is better). The best per column is highlighted in bold.

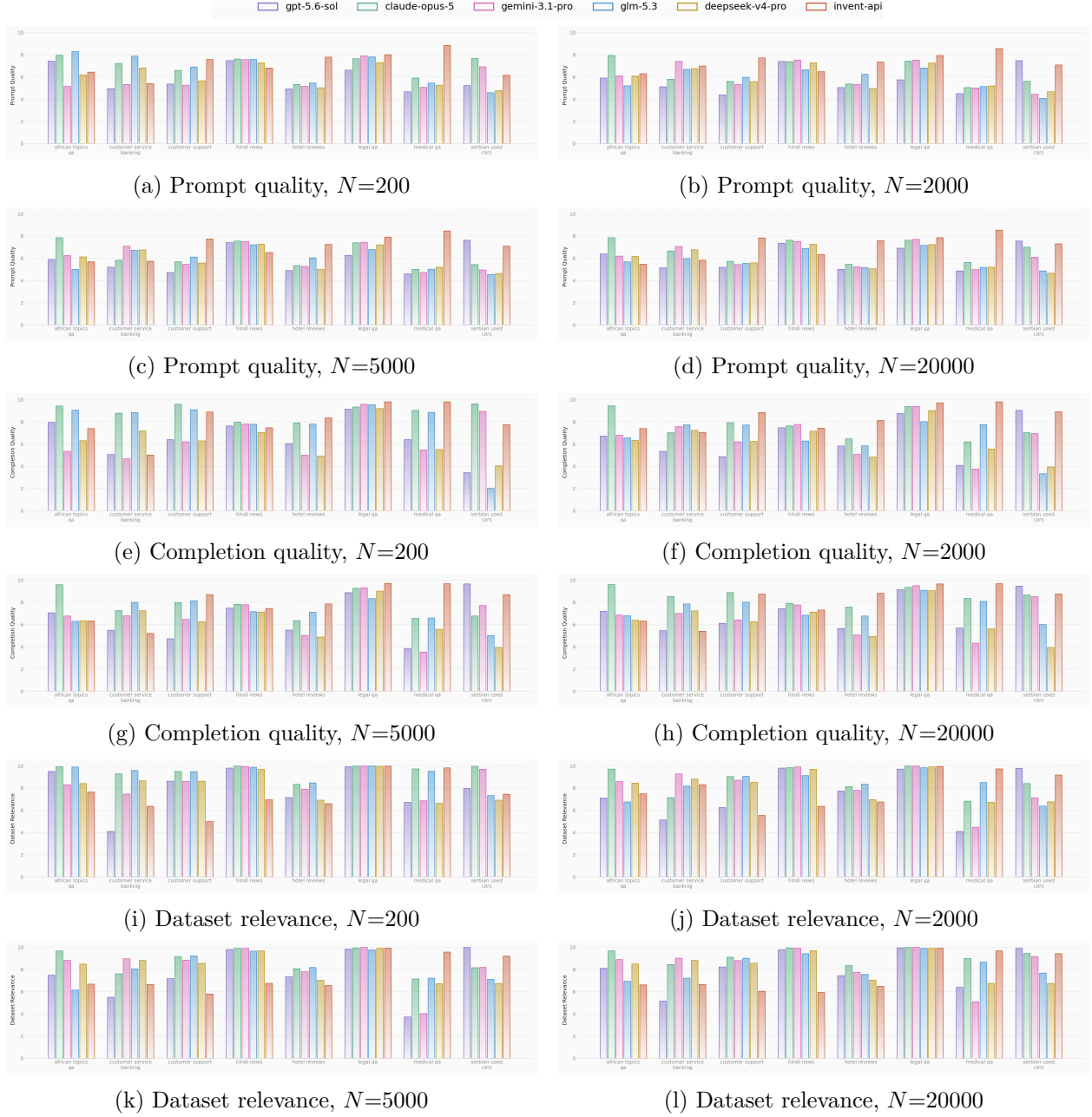


Figure 8: Per constrained dataset prompt quality, completion quality, and dataset relevance scores for each generator across dataset sizes $N \in \{200, 2000, 5000, 20000\}$.

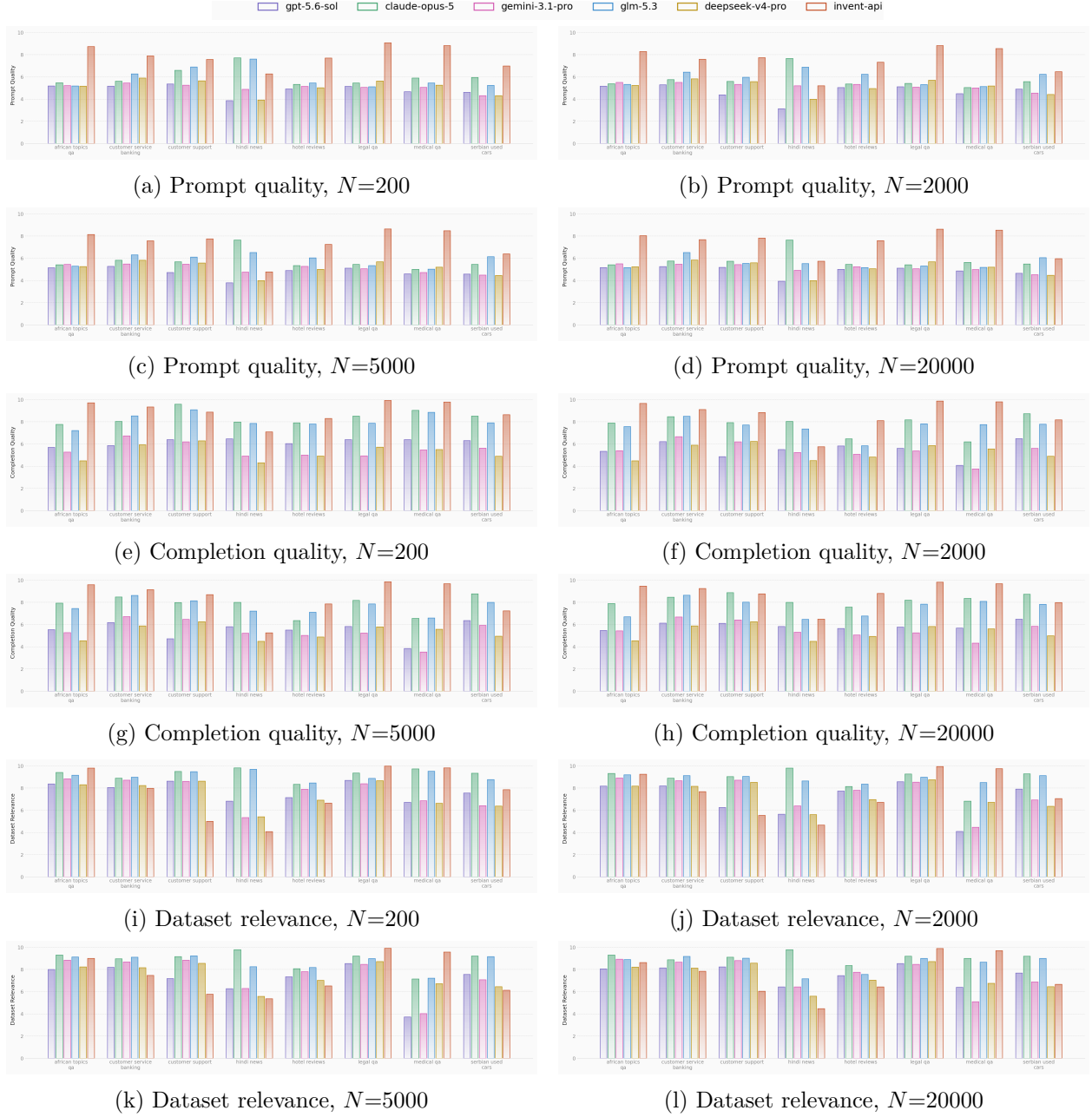


Figure 9: Per unconstrained dataset prompt quality, completion quality, and dataset relevance scores for each generator across dataset sizes $N \in \{200, 2000, 5000, 20000\}$.

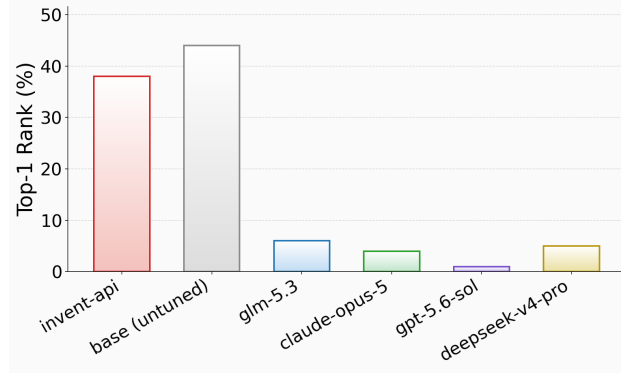


Figure 10: **Top-1 rank share for Qwen3.5-9B fine-tunes.** Percentage of held-out medical prompts on which each model ranks first (all six responses judged together per prompt); grey is the untuned base. Qwen3.5-9B has an unusually strong base that edges the Invent API fine-tune on both first-place rate (44% vs. 38%) and mean rank (2.17 vs. 2.28). Invent API fine-tuned variant nonetheless remains far ahead of all variants corresponding to other generators.

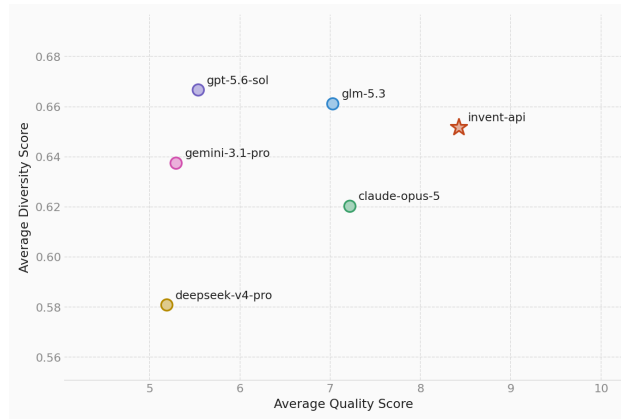


Figure 11: **Diversity vs. quality on 200-sample datasets.** Each point shows the average quality score (x-axis) and average diversity score, measured by DCScore ($\tau = 0.1$) (y-axis), across 8 dataset in unconstrained setting.

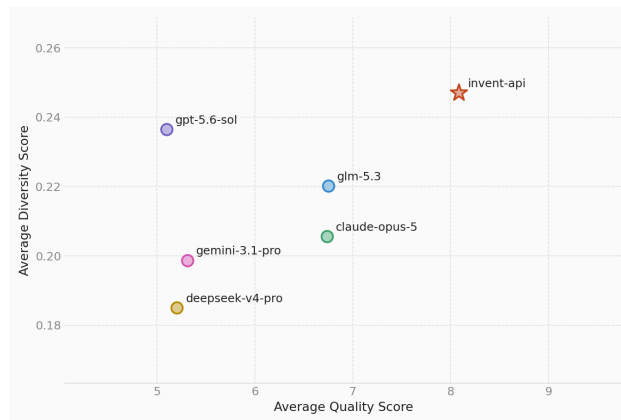


Figure 12: **Diversity vs. quality on 2000-sample datasets.** Each point shows the average quality score (x-axis) and average diversity score, measured by DCScore ($\tau = 0.1$) (y-axis), across 8 dataset in unconstrained setting.

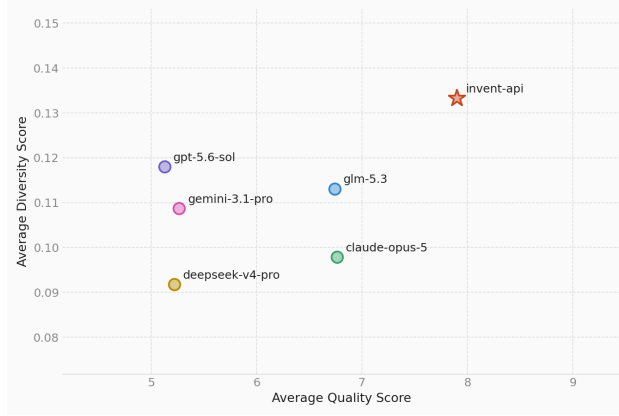


Figure 13: **Diversity vs. quality on 5000-sample datasets.** Each point shows the average quality score (x-axis) and average diversity score, measured by DCScore ($\tau = 0.1$) (y-axis), across 8 dataset in unconstrained setting.

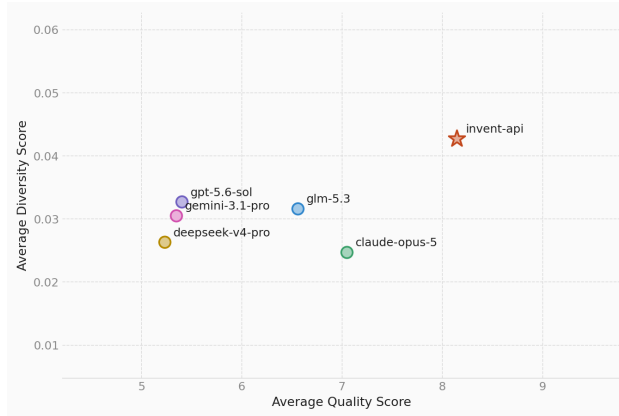


Figure 14: **Diversity vs. quality on 20000-sample datasets.** Each point shows the average quality score (x-axis) and average diversity score, measured by DCScore ($\tau = 0.1$) (y-axis), across 8 dataset in unconstrained setting.

A.8 Ranking Evaluation

This rubric compares models directly rather than scoring them one at a time. The models fine-tuned on the generated dataset and the un-tuned base model each answer the same held-out set of medical questions. For each question, the judge sees all responses in a single call and returns a best-to-worst ordering.

Ranking Prompt

Inputs

Below you are given a prompt from a user for an AI assistant, enclosed between `<prompt>` and `</prompt>`, and a set of responses from an AI assistant, enclosed between `<responses>` and `</responses>`.

`<prompt>{PROMPT}</prompt>`

`<responses>`

`{RESPONSES}`

`</responses>`

Evaluation Rubric

Your job is to evaluate and then rank the responses from best to worst. Please act as an impartial judge and evaluate the quality of the following `{N}` responses to the prompt. You will output two fields, “**ranking**” and “**ranking_explanation**”. “**ranking_explanation**” explains your rationale for why a completion is better than the others. “**ranking**” is a list of exactly `{N}` integers corresponding to the indices of the responses (the responses are zero-indexed), ranked from best to worst.

Evaluate each response based on the following criteria:

ACCURACY (Critical):

- Does the response provide factually correct information? Any hallucination or error should automatically result in a lower rank.
- Is every factual claim in the response correct?
- Does the response avoid hallucination, fabrication, or misleading implications?
- If a response contains any factual error, it must be ranked below all factually correct responses, regardless of how well it performs on other criteria.

INSTRUCTION FOLLOWING (Critical):

- Does the response directly answer the question asked — no more, no less?
- Does it address every explicit and implicit requirement of the prompt?
- Does it avoid answering a different or easier version of the question?
- Penalty: partial answers, redirections, or responses that technically answer but miss the core intent should be ranked lower.

COMPLETENESS VS. CONCISENESS (Medium Weight):

- Is the response as concise as possible without omitting necessary context?

- Does added detail genuinely improve usefulness, or is it padding?
- Penalty: unnecessary verbosity (adds noise, reduces clarity) and excessive brevity (omits context the user would reasonably need) should both be penalized relative to each other.

CLARITY & NATURALNESS (Low Weight — tiebreaker):

- Is the response grammatically correct and easy to understand?
- Does it read naturally, without awkward phrasing or redundancy?
- Use this criterion only to break ties when higher-priority criteria are equal across responses.

Act as an impartial judge. Do not favor any response based on length or style alone. If all responses are truly indistinguishable, rank by lowest index first.

Output Format

Your output should be A VALID JSON OBJECT (not in a code block), encoded in UTF-8, exactly matching the following JSON schema:

```
{
  "additionalProperties": false,
  "properties": {
    "ranking_explanation": {
      "description": "A brief rationale comparing the responses
                    and explaining the ranking order.",
      "title": "Ranking Explanation",
      "type": "string"
    },
    "ranking": {
      "description": "List of zero-indexed response indices
                    ordered from best to worst.",
      "items": { "type": "integer" },
      "title": "Ranking",
      "type": "array"
    }
  },
  "required": ["ranking_explanation", "ranking"],
  "title": "RankingExtractionOutput",
  "type": "object"
}
```

Your output should contain ONLY the JSON object and nothing else. **Do not** include explanations or any other text besides the JSON object.

A.9 Selected Dataset Samples

The subsequent table includes one sample including prompt and completion from each dataset produced by each generator.

Dataset	Prompt	Completion
Model: Claude Opus 5		
General Customer Support	“How many people does the 8-inch cake serve?”	“Our 8-inch round cakes serve 10-12 people with standard party-sized slices, or up to 16 if you’re cutting smaller dessert portions alongside other sweets. If you’re expecting more than 15 guests as the main dessert, I’d suggest the 10-inch (serves 16-20) or two 8-inch cakes in different flavours, which is popular for b [...]”
Hotel Reviews	“Write a five-star review of a hotel that felt like great value for money.”	“Five stars purely on value. Ninety-four dollars a night in a major city for a clean, quiet room with a good bed, a real breakfast included, and staff who acted like they were glad I was there. No resort fee, no parking gouge, no surprise charges. The furniture is a decade old and the lobby is plain, and I could not car [...]”
Customer Service Banking	“Nobody told me I needed a PIN to activate. This whole thing has wasted my morning.”	“I’m very sorry that wasn’t made clear to you, and I appreciate you taking the time to explain what happened. Let’s make the rest of this quick. I can activate the card and set up a temporary PIN with you right now, which you can change at any ATM. Could you confirm the last four digits of your card and your mailing ZIP [...]”
Hindi News	पाकिस्तान को हराकर भारत ने विश्व कप में बनाए रखा अजेय रिकॉर्ड	अहमदाबाद के नरेंद्र मोदी स्टेडियम में एक लाख से ज्यादा दर्शकों के सामने भारत ने पाकिस्तान को सात विकेट से हरा दिया। जसप्रीत बुमराह और मोहम्मद सिराज की घातक गेंदबाजी से पाकिस्तान 191 रन पर ढेर हो गया। [...]
Legal QA	“In the United States, how long does copyright protection generally last for a work created by an individual author after 1977? A) 25 years from creation B) The life of the author plus 70 years C) 50 years from first publication D) Forever, [...]”	“B”

Dataset	Prompt	Completion
Serbian Car Ads	“VW Caddy 1.6 TDI, 2012. godište, teretni, 301.000 km, korišćen za dostavu. Redovno servisiran na svakih 15.000 km, nedavno zamenjena turbina i kvačilo. Ima klimu, ABS, servo volan, radio i pregradnu mrežu.”	<code>{"vehicle_description": "Volkswagen Caddy teretni iz 2012. godine sa 1.6 TDI motorom i pređenih 301.000 km. Vozilo je korišćeno za dostavu.", "service_history": "Redovno servisiran na svakih 15.000 km; nedavno zamenjeni turbina i kvačilo.", "equipment_packages": ["klima uređaj", "ABS", "servo volan", "radio", "pregradn [...]</code>
African QA	“system: You are a student sitting an oral examination in African literature. Answer the examiner’s questions accurately and concisely. user: Let us begin with Chinua Achebe. When was Things Fall Apart published, and in what setting? ass [...]”	“The novel is built in three parts. The first two establish Igbo society on its own terms: its proverbs, festivals, legal procedures, religion and internal tensions, including Okonkwo’s rigidity and the killing of Ikemefuna. Only in the third part does the missionary and colonial administration arrive. By giving roughly [...]”
Model: Gemini 3.1 Pro		
General Customer Support	“How tall is the BaristaPro? I need to know if it fits under my cabinets.”	“The BaristaPro is 14.5 inches tall. Most standard kitchen cabinets have an 18-inch clearance, so it should fit perfectly.”
Hotel Reviews	“Provide a review with great service and location, but bad room and facilities.”	“Loved being right on the beach, and the valets were so polite. However, the bed was completely caved in and there was no Wi-Fi.”
Customer Service Banking	“I am stuck in a password reset loop just trying to activate this stupid card.”	“Welcome to our support team! I am incredibly sorry for the frustrating password loop. Let us skip the website for now; I would be delighted to verify you and activate your card directly.”
Hindi News	भारतीय पुरुष हॉकी टीम ने एशियन गेम्स में जीता स्वर्ण पदक	हरमनप्रीत सिंह की कप्तानी में भारतीय पुरुष हॉकी टीम ने जापान को 5-1 से हराकर एशियाई खेलों में स्वर्ण पदक जीता और पेरिस ओलंपिक के लिए सीधे क्वालीफाई किया।
Legal QA	“What is the general duration of copyright for a work created by an individual author in the United States today? A) 50 years B) Life of the author plus 70 years C) 95 years from publication D) 120 years from creation”	“B”

Dataset	Prompt	Completion
Serbian Car Ads	“Na prodaju VW Golf 5, 2007. godište, u odličnom stanju. Redovno održavan u ovlašćenom servisu. Od opreme poseduje automatsku klimu, tempomat i alu felne.”	<code>{"vehicle_description": "VW Golf 5, 2007. godište, u odličnom stanju.", "service_history": "Redovno održavan u ovlašćenom servisu.", "equipment_packages": ["automatska klima", "tempomat", "alu felne"]}</code>
African QA	“user: Do we memorize publication dates for Achebe? assistant: Just know the general era it was written. user: So exact years aren't needed?”	“No, focus on the pre-colonial setting instead of publication years.”
Model: GPT 5.6 So1		
General Customer Support	“Does the ViewMax monitor include an HDMI cable?”	“Yes, the box includes one HDMI cable, a power adapter, the stand, and setup documentation. A DisplayPort cable is sold separately.”
Hotel Reviews	“Write a balanced hotel review mentioning friendly staff, a convenient location, and aging amenities.”	“The staff were genuinely friendly, and the location near shops and transit was extremely convenient. Our room was clean and comfortable, but the aging elevator, worn gym equipment, and dated bathroom need attention. Overall, it was a decent stay with room for improvement.”
Customer Service Banking	“My new card has no activation sticker. What should I do?”	“I'll be happy to help. Sign in through the official mobile app or online banking site and look for “Activate card.” You can also call the verified customer service number printed on the back of the card.”
Hindi News	रोहित शर्मा ने छक्कों का नया रिकॉर्ड बनाया	भारतीय कप्तान रोहित शर्मा ने आक्रामक पारी के दौरान अंतरराष्ट्रीय क्रिकेट में छक्कों का नया कीर्तिमान स्थापित किया। उनकी बल्लेबाजी ने भारत को मजबूत शुरुआत दिलाई।
Legal QA	“1. Which protection generally applies automatically when an original poem is fixed in writing? A. Copyright B. Patent C. Trademark D. Trade secret”	“A”

Dataset	Prompt	Completion
Serbian Car Ads	“Volkswagen Golf 7 1.6 TDI, 2016. godište, 178.000 km. Uvezen iz Nemačke, bez ulaganja, redovno održavan. Comfortline paket, navigacija, tempomat i parking senzori.”	<code>{"vehicle_description": "Golf 7 sa dizel motorom 1.6 TDI, 2016. godište i 178.000 km. Vozilo je uvezeno iz Nemačke, očuvano i spremno za vožnju bez dodatnih ulaganja.", "service_history": "Redovno servisiran; dostupna servisna dokumentacija iz Nemačke.", "equipment_packages": ["Comfortline", "Navigacija", "Tempomat", " [...]</code>
African QA	“system: You are a student taking an exam on African history. user: Teacher: What evidence shows that Great Zimbabwe was a major regional center? assistant: Student: Its stone architecture and imported goods indicate wealth and trade. [...]”	“Archaeologists found glass beads, Chinese ceramics, and coins linked to Indian Ocean commerce. Together with local gold-working evidence and monumental dry-stone walls, these finds show that Great Zimbabwe controlled resources and participated in long-distance trade rather than existing as an isolated settlement.”
Model: GLM-5.3		
General Customer Support	“Can you tell me if the SmartFit watch is waterproof? I want to use it for swimming.”	“Great question! The SmartFit watch is water-resistant up to 50 meters (5 ATM), which means it’s suitable for swimming, showering, and shallow water activities. However, we don’t recommend it for diving or high-velocity water sports. After swimming, we suggest rinsing it with fresh water to remove chlorine or salt resid [...]”
Hotel Reviews	“Write a mixed hotel review: great service but a noisy room at The Aveline.”	“The Aveline deserves praise and a complaint in the same breath. The staff were uniformly excellent, from the flawless check-in to the concierge who arranged our anniversary dinner with a hand-delivered card. But our room faced the street, and the noise was constant: delivery trucks, nightlife crowds, and a persistent m [...]”

Dataset	Prompt	Completion
Customer Service Banking	“My husband’s card came and I want to activate it for him since he’s deployed. Why is this so difficult?”	“First, thank you so much for your family’s service, and I’m so sorry this has been difficult for you while your husband is away. That’s an added stress you simply don’t need. For security purposes, primary cardholders do need to be verified, but we have special procedures for military families. May I ask if your husband [...]”
Hindi News	□20 विश्व कप फाइनल: 17 साल बाद भारत फिर बना विश्व चैंपियन, दक्षिण अफ्रीका को 7 रन से हराया	बारबाडोस के केंसिंग्टन ओवल में भारत ने दक्षिण अफ्रीका को 7 रन से हराकर □20 विश्व कप अपने नाम किया। विराट कोहली की 76 रन की शानदार पारी और आखिरी ओवरों में हार्दिक पांड्या की गेंदबाजी ने टीम को यह यादगा [...]
Legal QA	“In the United States, what is the duration of copyright protection for a work created by an individual author on or after January 1, 1978? A. 28 years from creation with the option to renew B. The life of the author plus 70 years C. 95 year [...]”	“B”
Serbian Car Ads	“Toyota Yaris, 2012. godište, 1.0, 69 ks, 130.000 km. Gradski auto sa automatskom klimom, ekstremno ekonomičan. Servisiran u ovlašćenom servisu, knjižica kompletna.”	{“vehicle_description”: “Toyota Yaris III, hečbek, 2012. godište, benzinski motor 1.0 VVT-i od 69 ks, prešao 130.000 km, bela boja, manuelni menjač sa 5 stepeni, prednji pogon, pouzdan gradski automobil”, “service_history”: “Kompletna servisna knjižica iz ovlašćenog servisa Toyote, ulje i filteri menjani na 126.000 km, [...]”
African QA	“system: You are Professor Amara Diallo, a rigorous but encouraging teacher of African history conducting an oral exam. Give precise feedback on each answer, correct errors kindly, and end your turns with a follow-up question. user: Good m [...]”	“Excellent. The pilgrimage also drew scholars, architects, and merchants toward Mali, boosting Timbuktu’s growth as a center of learning, and Mali appeared on European maps such as the Catalan Atlas of 1375. Here is a subtler question: why did Timbuktu flourish as a scholarly city, and what recent event in Mali threaten [...]”
Model: Invent API		

Dataset	Prompt	Completion
General Customer Support	“Customer: ‘My smart watch is not syncing heart rate data to the health app during workouts, even though the step counter works. I’ve unpaired and repaired the device three times. Does this sound like a sensor hardware failure or a software [...]”	“Hello! I understand how frustrating it is when your health data isn’t syncing, especially after you’ve already tried unpairing and repairing multiple times. Since your step counter is working, we know the Bluetooth connection is active, but we need to determine if the heart rate sensor itself is failing or if the data [...]”
Hotel Reviews	“Write a two-sentence hotel assessment covering service, room quality, location, and amenities that balances both positive and negative points.”	“The concierge service was impeccable and the room spacious, yet the housekeeping schedule was inconsistent and the Wi-Fi unreliable. While the location offers unbeatable access to the city center, the lack of a functional gym and a noisy pool area significantly detracts from the overall experience.”
Customer Service Banking	“Hi, I’ve been on hold for twenty minutes trying to activate my new credit card, and now the automated system is telling me it’s already blocked! This is incredibly frustrating. Can you please help me sort this out immediately?”	“Hello! I completely understand your frustration, and I sincerely apologize for the long wait and the confusion you’ve experienced with our automated system. My name is Sarah, and I’m here to help you get this resolved right away. Let’s take a look at your account together. Could you please verify your full name and the [...]”
Hindi News	भारत ने ऑस्ट्रेलिया को हराकर टी-20 विश्व कप का खिताब जीता	रोमांचक फाइनल मुकाबले में भारतीय क्रिकेट टीम ने ऑस्ट्रेलिया को 6 विकेट से हराकर टी-20 विश्व कप का खिताब अपने नाम किया। ऑस्ट्रेलिया ने पहले बल्लेबाजी करते हुए निर्धारित 20 ओवर में 184 रन बनाए थे, जिसके [...]
Legal QA	“Hypothetically, if the U.S. Supreme Court were to rule on a case involving the statutory interpretation of banking fee disclosures with the same ironic detachment observed in certain 2012 dissenting opinions, which rhetorical approach would [...]”	“C”

Dataset	Prompt	Completion
Serbian Car Ads	“Sumiraj oglas za polovni automobil: Renault Megane iz 2015. godine, prešao 145.000 km. U opisu stoji da je redovno održavan, sa velikim servisom urađenim na 120.000 km, i da poseduje klimu, navigaciju i kožna sedišta.”	{ "vehicle_description": "Renault Megane, godište 2015, pređeno 145.000 km. Vozilo je u dobrom stanju, redovno održavano.", "service_history": "Redovno servisiranje dokumentovano; veliki servis izvršen na 120.000 km.", "equipment_packages": ["Klima uređaj", "Navigacioni sistem", "Kožna sedišta"] }
African QA	“system: You are an examiner for a course on Pre-Colonial African Civilizations. The student is taking a comprehensive oral exam. user: Teacher: Let’s discuss the Aksumite Empire. Analyze the transition from polytheism to Christianity in t [...]”	“The adoption of Christianity strengthened Aksum’s alignment with the Byzantine Empire, its primary Mediterranean trading partner, facilitating smoother commercial and diplomatic relations. However, this religious shift created a growing ideological rift with the Himyarite Kingdom in Southern Arabia. While their relatio [...]”